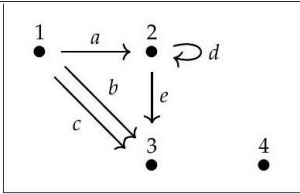
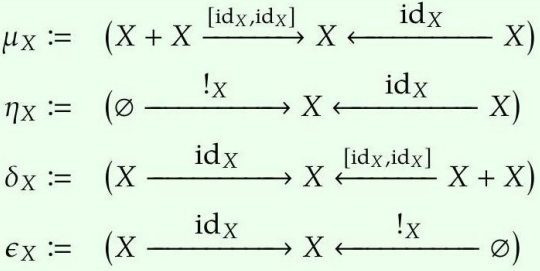
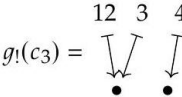
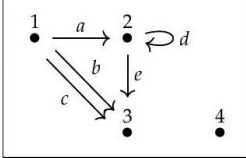
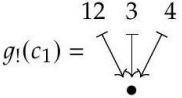
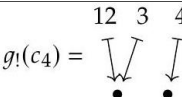


fong-spivak-seven-sketches: equation evidence

Display equations

Identifier	Page	Conf.	LaTeX source	Rendered	Image
fong-spivak-seven-sketches-EQ0061 (3.61) [conf 0.008]	110	■0.008	I=\left[\begin{array}{ccc} 1 & \underset{\bullet}{a} & \overset{2}{\bullet} \\ & \downarrow & \\ & \downarrow & \\ & \downarrow & \bullet \end{array} \right] \begin{array}{l} b \\ d \end{array}	$I = \left[\begin{array}{ccc} 1 & \overset{2}{\bullet} & b \\ & \downarrow & \\ & \downarrow & \\ & \downarrow & \bullet \end{array} \right] \begin{array}{l} d \\ \end{array}$	
fong-spivak-seven-sketches-EQ0063 [conf 0.014]	111	■0.014	G:=\stackrel{1}{\bullet} \xrightarrow{a} \stackrel{2}{\bullet} \xrightarrow{b} \stackrel{3}{\bullet} \quad H:=\stackrel{4}{\bullet} \xrightarrow[c]{d} \stackrel{5}{\bullet} \xrightarrow{e} \text{Pe}	$G := \overset{1}{\bullet} \xrightarrow{a} \overset{2}{\bullet} \xrightarrow{b} \overset{3}{\bullet} \quad H := \overset{4}{\bullet} \xrightarrow[c]{d} \overset{5}{\bullet} \xrightarrow{e} \text{Pe}$	
fong-spivak-seven-sketches-EQ0133 [conf 0.025]	214	■0.025	\begin{aligned} \mu_X &:= \left(X + X \xrightarrow{[\text{id}_X, \text{id}_X]} X \xleftarrow{\text{id}_X} X \right) \\ \eta_X &:= \left(\emptyset \xrightarrow{!_X} X \xleftarrow{\text{id}_X} X \right) \\ \delta_X &:= \left(X \xrightarrow{\text{id}_X} X \xleftarrow{[\text{id}_X, \text{id}_X]} X + X \right) \\ \epsilon_X &:= \left(X \xrightarrow{\text{id}_X} X \xleftarrow{!_X} \emptyset \right) \end{aligned}	$\mu_X := \left(X + X \xrightarrow{[\text{id}_X, \text{id}_X]} X \xleftrightarrow{\text{id}_X} X \right)$ $\eta_X := \left(\emptyset \xrightarrow{!_X} X \xleftrightarrow{\text{id}_X} X \right)$ $\delta_X := \left(X \xrightarrow{\text{id}_X} X \xrightarrow{[\text{id}_X, \text{id}_X]} X + X \right)$ $\epsilon_X := \left(X \xrightarrow{\text{id}_X} X \xleftrightarrow{!_X} \emptyset \right)$	
fong-spivak-seven-sketches-EQ0188 [conf 0.038]	279	■0.038	g_!(c_3)=\overset{12}{\downarrow} \overset{3}{\downarrow} \overset{4}{\bullet}	$g_!(c_3) = \overset{12}{\downarrow} \overset{3}{\downarrow} \overset{4}{\bullet}$	
fong-spivak-seven-sketches-EQ0181 [conf 0.044]	274	■0.044	G=\left[\begin{array}{ccc} 1 & \underset{\bullet}{a} & \overset{2}{\bullet} \\ & \downarrow & \downarrow \\ & \downarrow & \\ & \downarrow & \\ & \downarrow & \end{array} \right]	$G = \left[\begin{array}{ccc} 1 & \overset{2}{\bullet} & \\ & \downarrow & \downarrow \\ & \downarrow & \\ & \downarrow & \end{array} \right]$	
fong-spivak-seven-sketches-EQ0186 [conf 0.050]	279	■0.050	g_!(c_1)=\overset{12}{\downarrow} \overset{3}{\downarrow} \overset{4}{\bullet}	$g_!(c_1) = \overset{12}{\downarrow} \overset{3}{\downarrow} \overset{4}{\bullet}$	
fong-spivak-seven-sketches-EQ0189	279	■0.108	g_!(c_4)=\Pi \overset{12}{\downarrow} \overset{3}{\downarrow} \overset{4}{\bullet}	$g_!(c_4) = \Pi \overset{12}{\downarrow} \overset{3}{\downarrow} \overset{4}{\bullet}$	

Identifier	Page	Conf.	LaTeX source	Rendered	Image
fong-spivak-seven-sketches_1 EQ0073 (3.77)	118	■ 0.114	$\mathcal{G}:=\left \stackrel{\text{Email}}{\underset{\text{received_by}}{\stackrel{\text{sent_by}}{\rightarrow}}}\right ^{\text{Address}}\bullet$	(not rendered)	$\mathcal{G} :=$ Email sent_by Address received_by no equations
fong-spivak-seven-sketches_1 EQ0245	311	■ 0.129	$a_{\{m\}}:=(1+\cdots+1+\cdots) ; \cdots ;(1+1+\supset-);(1+\cdots) ; \cdots: m \rightarrow 1$	$a_m := (1 + \cdots + 1 + \cdots); \cdots; (1 + 1 + \supset -); (1 + \cdots); \cdots: m \rightarrow 1$	$a_m := (1 + \cdots + 1 + \supset -) \circ \cdots \circ (1 + 1 + \supset -) \circ (1 + \supset -) \supset -: m \rightarrow 1$
fong-spivak-seven-sketches_1 EQ0083	136	■ 0.147	$\text{Chassis: } (\text{load} \times \text{velocity}) \rightarrow (\text{torque} \times \text{speed} \times \$)$	Chassis: (load × velocity) → (torque × speed × \$)	Chassis: (load × velocity) → (torque × speed × \$)
fong-spivak-seven-sketches_1 EQ0205	290	■ 0.147	$A, \quad A \div f, \quad A \div g, \quad A \div f \div h, \quad A \div g \mid i, \quad B, \quad B \div h, \quad C, \quad C : i, \quad D$	$A, \quad A \div f, \quad A \div g, \quad A \div f \div h, \quad A \div g \mid i, \quad B, \quad B \div h, \quad C, \quad C : i, \quad D$	$A, \quad A \circ f, \quad A \circ g, \quad A \circ f \circ h, \quad A \circ g \circ i, \quad B, \quad B \circ h, \quad C, \quad C \circ i, \quad D$
fong-spivak-seven-sketches_1 EQ0064 (3.65)	112	■ 0.147	$\mathbf{Gr}:=\stackrel{\text{Arrow}}{\rightarrow}\stackrel{\text{source}}{\Rightarrow}\stackrel{\text{Vertex}}{\text{target}}$	$\mathbf{Gr} := \stackrel{\text{Arrow}}{\rightarrow} \stackrel{\text{source}}{\Rightarrow} \stackrel{\text{Vertex}}{\text{target}}$	Arrow source Vertex target no equations State next no equations
fong-spivak-seven-sketches_1 EQ0214	294	■ 0.168	$H:=\stackrel{4}{\rightarrow}\stackrel{d}{\rightarrow}\stackrel{5}{\rightarrow}\bullet>e$	$H := \stackrel{4}{\rightarrow} \stackrel{d}{\rightarrow} \stackrel{5}{\rightarrow} \bullet > e$	$H := \left \begin{array}{c} 4 \xrightarrow[d]{d} 5 \end{array} \right \bullet \curvearrowright e$
fong-spivak-seven-sketches_1 EQ0115	182	■ 0.203	$G_{\{R\}}:=\{\supset-, \circ-, \subset, \longrightarrow\} \cup \{-a- \mid a \in R\},$	$G_R := \{ \supset -, \circ -, \subset, \longrightarrow \} \cup \{ -a - \mid a \in R \},$	$G_R := \left\{ \supset -, \circ -, \subset, \longrightarrow \right\} \cup \left\{ -\Box a - \mid a \in R \right\},$
fong-spivak-seven-sketches_1 EQ0187	279	■ 0.215	$g_{\{!\}\left(c_{\{2\}}\right)}=\Psi_{\bullet}^{12} J^{34}$	$g_{\{!\}\left(c_{\{2\}}\right)} = \Psi_{\bullet}^{12} J^{34}$	$g_{\{!\}\left(c_{\{2\}}\right)} = \begin{array}{c} 12 \quad 3 \quad 4 \\ \downarrow \quad \downarrow \\ \bullet \quad \bullet \end{array}$
fong-spivak-seven-sketches_1 EQ0094	148	■ 0.259	$\begin{array}{l} S \times (T \times U) := \{(s, (t, u)) \mid s \in S, t \in T, u \in U\} \\ \text{mid } s \in S, t \in T, u \in U \} \stackrel{?}{=} (S \times T) \times U := \{((s, t), u) \mid s \in S, t \in T, u \in U\}. \end{array}$	$S \times (T \times U) := \{(s, (t, u)) \mid s \in S, t \in T, u \in U\} \stackrel{?}{=} (S \times T) \times U := \{((s, t), u) \mid s \in S, t \in T, u \in U\}.$	$S \times (T \times U) := \{(s, (t, u)) \mid s \in S, t \in T, u \in U\} \stackrel{?}{=} (S \times T) \times U := \{((s, t), u) \mid s \in S, t \in T, u \in U\}.$
fong-spivak-seven-sketches_1 EQ0072	118	■ 0.263	$\Sigma_{!}: \mathcal{C} - \mathcal{I} \setminus \sqcup \rightarrow \mathbf{Set} \quad \text{and} \quad \Pi_{!}: \mathcal{C} - \mathbf{Inst} \rightarrow \mathbf{Set}.$	$\Sigma_{!}: \mathcal{C} - \mathcal{I} \setminus \sqcup \rightarrow \mathbf{Set} \quad \text{and} \quad \Pi_{!}: \mathcal{C} - \mathbf{Inst} \rightarrow \mathbf{Set}.$	$\Sigma_{!}: \mathcal{C}\text{-}\mathbf{Inst} \rightarrow \mathbf{Set} \quad \text{and} \quad \Pi_{!}: \mathcal{C}\text{-}\mathbf{Inst} \rightarrow \mathbf{Set}.$
fong-spivak-seven-sketches_1 EQ0020 (2.10)	55	■ 0.266	$\begin{array}{l} h_{\{1\}}:=\{2 \diamond, 3 \diamond, 10 \spadesuit, J \spadesuit, Q \spadesuit\} \\ \bullet, J \bullet, Q \bullet \} \text{ \& } i_{\{1\}}:=\{4 \diamond, 4 \diamond, 6 \diamond, 6 \diamond, 10 \diamond\} \\ h_{\{2\}}:=\{2 \diamond, 3 \diamond, 4 \diamond, K \star, A \diamond\} \\ \text{ \& } i_{\{2\}}:=\{5 \diamond, 5 \diamond, 7 \diamond, J \diamond, Q \diamond\}. \end{array}$	$h_1 := \{2 \diamond, 3 \diamond, 10 \spadesuit, J \spadesuit, Q \spadesuit\} \quad i_1 := \{4 \diamond, 4 \diamond, 6 \diamond, 6 \diamond, 10 \diamond\}$ $h_2 := \{2 \diamond, 3 \diamond, 4 \diamond, K \star, A \diamond\} \quad i_2 := \{5 \diamond, 5 \diamond, 7 \diamond, J \diamond, Q \diamond\}.$	$h_1 := \{2 \heartsuit, 3 \heartsuit, 10 \spadesuit, J \spadesuit, Q \spadesuit\} \quad i_1 := \{4 \clubsuit, 4 \spadesuit, 6 \heartsuit, 6 \diamond, 10 \diamond\}$ $h_2 := \{2 \diamond, 3 \diamond, 4 \diamond, K \spadesuit, A \spadesuit\} \quad i_2 := \{5 \spadesuit, 5 \heartsuit, 7 \heartsuit, J \diamond, Q \diamond\}.$
fong-spivak-seven-sketches_1 EQ0033	73	■ 0.268	$X:=\operatorname{Ob}\{0b\}(\mathfrak{X}) \quad \text{quad} \quad d(x, y):=\mathfrak{X}(x, y) \text{ .}$	$X := \operatorname{Ob}(\mathfrak{X}) \quad d(x, y) := \mathfrak{X}(x, y).$	$X := \operatorname{Ob}(\mathfrak{X}) \quad d(x, y) := \mathfrak{X}(x, y).$

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fong-spivak-seven-sketches_7 EQ0175 (7.82)	268	■ 0.278	$\begin{aligned} & \forall (t : \mathbb{R}). \forall (t : \mathbb{R}). \text{bad_pos}(D) \Rightarrow \\ & \left(\text{bad_pos}(D) \Rightarrow \exists (r : \mathbb{R}). (0 < r < 1) \wedge \forall (r' : \mathbb{R}). 0 \leq r' \leq 5 \Rightarrow \text{engaged}(T) \right) . \\ & \end{aligned}$	$\forall (t : \mathbb{R}). \text{bad_pos}(D) \Rightarrow \exists (r : \mathbb{R}). (0 < r < 1) \wedge \forall (r' : \mathbb{R}). 0 \leq r' \leq 5 \Rightarrow \text{engaged}(T).$	$\forall (t : \mathbb{R}). \text{bad_pos}(D) \Rightarrow \exists (r : \mathbb{R}). (0 < r < 1) \wedge \forall (r' : \mathbb{R}). 0 \leq r' \leq 5 \Rightarrow \text{engaged}(T).$
fong-spivak-seven-sketches_7 EQ0057	97	■ 0.293	$\mathcal{D} := \left[\begin{array}{c} s \\ \hat{L} \\ b \\ z \\ sisjs \text{ is } s = s \text{ is} \end{array} \right]$	$\mathcal{D} := \left[\begin{array}{c} s \\ \hat{L} \\ b \\ z \\ sisjs \text{ is } s = s \text{ is} \end{array} \right]$	$\mathcal{D} := \left[\begin{array}{c} s \\ \hat{L} \\ b \\ z \\ sisjs \text{ is } s = s \text{ is} \end{array} \right]$
fong-spivak-seven-sketches_7 EQ0184	274	■ 0.304	$\begin{array}{l} (\bullet)(\circ)(*) \leq (\bullet)(\circ)(*) \quad (\bullet)(\circ)(*) \leq (\bullet\circ)(*) \\ (\bullet)(\circ)(*) \leq (\bullet)(\circ*) \quad (\bullet)(\circ)(*) \leq (\bullet\circ*) \\ (\bullet\circ)(*) \leq (\bullet\circ*) \quad (\bullet*)(\circ) \leq (\bullet*)(\circ) \\ (\bullet)(\circ*) \leq (\bullet)(\circ*) \quad (\bullet)(\circ*) \leq (\bullet\circ*) \\ (\bullet\circ)(*) \leq (\bullet\circ*) \quad (\bullet*)(\circ) \leq (\bullet*)(\circ) \\ (\bullet*)(\circ) \leq (\bullet\circ*) \quad (\bullet\circ*) \leq (\bullet\circ*) \\ (\bullet\circ*) \leq (\bullet\circ*) \end{array}$	$\begin{array}{l} (\bullet)(\circ)(*) \leq (\bullet)(\circ)(*) \quad (\bullet)(\circ)(*) \leq (\bullet\circ)(*) \\ (\bullet)(\circ)(*) \leq (\bullet)(\circ*) \quad (\bullet)(\circ)(*) \leq (\bullet\circ*) \\ (\bullet\circ)(*) \leq (\bullet\circ*) \quad (\bullet*)(\circ) \leq (\bullet*)(\circ) \\ (\bullet)(\circ*) \leq (\bullet)(\circ*) \quad (\bullet)(\circ*) \leq (\bullet\circ*) \\ (\bullet\circ)(*) \leq (\bullet\circ*) \quad (\bullet*)(\circ) \leq (\bullet*)(\circ) \\ (\bullet*)(\circ) \leq (\bullet\circ*) \quad (\bullet\circ*) \leq (\bullet\circ*) \\ (\bullet\circ*) \leq (\bullet\circ*) \end{array}$	$\begin{array}{l} (\bullet)(\circ)(*) \leq (\bullet)(\circ)(*) \quad (\bullet)(\circ)(*) \leq (\bullet\circ)(*) \\ (\bullet)(\circ)(*) \leq (\bullet)(\circ*) \quad (\bullet)(\circ)(*) \leq (\bullet\circ*) \\ (\bullet\circ)(*) \leq (\bullet\circ*) \quad (\bullet*)(\circ) \leq (\bullet*)(\circ) \\ (\bullet)(\circ*) \leq (\bullet)(\circ*) \quad (\bullet)(\circ*) \leq (\bullet\circ*) \\ (\bullet\circ)(*) \leq (\bullet\circ*) \quad (\bullet*)(\circ) \leq (\bullet*)(\circ) \\ (\bullet*)(\circ) \leq (\bullet\circ*) \quad (\bullet\circ*) \leq (\bullet\circ*) \\ (\bullet\circ*) \leq (\bullet\circ*) \end{array}$
fong-spivak-seven-sketches_7 EQ0056	97	■ 0.330	$e := \begin{array}{ c } \hline s \\ \mathcal{Q} \\ z \\ s; s = z \\ \hline \end{array}$	$e := \begin{array}{ c } \hline s \\ \mathcal{Q} \\ z \\ s; s = z \\ \hline \end{array}$	$\mathcal{C} := \begin{array}{ c } \hline s \\ \mathcal{Q} \\ z \\ s \circ s = z \\ \hline \end{array}$
fong-spivak-seven-sketches_7 EQ0109 (5.47)	175	■ 0.331	$G_{\{R\}} := \{\supset, \circ, \subset, \longrightarrow\} \cup \{-a- \mid a \in R\},$	$G_R := \{\supset, \circ, \subset, \longrightarrow\} \cup \{-a- \mid a \in R\},$	$G_R := \{\supset, \circ, \subset, \longrightarrow\} \cup \{-a- \mid a \in R\},$
fong-spivak-seven-sketches_7 EQ0059 (3.47)	106	■ 0.333	$C := \begin{array}{ c } \hline s \\ \mathcal{Q} \\ b \\ s; s = s \\ \hline \end{array}$	$C := \begin{array}{ c } \hline s \\ \mathcal{Q} \\ b \\ s; s = s \\ \hline \end{array}$	$\mathcal{C} := \begin{array}{ c } \hline s \\ \mathcal{Q} \\ z \\ s \circ s = s \\ \hline \end{array}$
fong-spivak-seven-sketches_7 EQ0113	178	■ 0.335	$\mapsto(1) \quad \text{and} \quad X \mapsto \left(\begin{array}{cc} 0 & 1 \\ 1 & 0 \end{array} \right) \quad \text{and} \quad -a- \mapsto (a)$	$\mapsto(1) \quad \text{and} \quad X \mapsto \left(\begin{array}{cc} 0 & 1 \\ 1 & 0 \end{array} \right) \quad \text{and} \quad -a- \mapsto (a)$	$\mapsto(1) \quad \text{and} \quad X \mapsto \left(\begin{array}{cc} 0 & 1 \\ 1 & 0 \end{array} \right) \quad \text{and} \quad -a- \mapsto (a)$
fong-spivak-seven-sketches_7 EQ0226	303	■ 0.335	$\mathcal{X}(x, z) = \mathcal{X}(x, z) \otimes I \leq \mathcal{X}(x, z) \otimes \mathcal{X}(z, z) \leq \bigvee_{y \in \mathcal{X}} \mathcal{X}(x, y) \otimes \mathcal{X}(y, z),$	$\mathcal{X}(x, z) = \mathcal{X}(x, z) \otimes I \leq \mathcal{X}(x, z) \otimes \mathcal{X}(z, z) \leq \bigvee_{y \in \mathcal{X}} \mathcal{X}(x, y) \otimes \mathcal{X}(y, z),$	$\mathcal{X}(x, z) = \mathcal{X}(x, z) \otimes I \leq \mathcal{X}(x, z) \otimes \mathcal{X}(z, z) \leq \bigvee_{y \in \mathcal{X}} \mathcal{X}(x, y) \otimes \mathcal{X}(y, z),$

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fong-spivak-seven-sketches EQ0054 (3.11)	95	0.526	$\mathbf{3} := \operatorname{Free} \left(\left(\begin{array}{c} v_1 \\ \bullet \end{array} \right) \xrightarrow{f_1} \left(\begin{array}{c} v_2 \\ \bullet \end{array} \right) \xrightarrow{f_2} \left(\begin{array}{c} v_3 \\ \bullet \end{array} \right) \right)$	$3 := \operatorname{Free} \left(\begin{array}{c} v_1 \\ \bullet \end{array} \xrightarrow{f_1} \begin{array}{c} v_2 \\ \bullet \end{array} \xrightarrow{f_2} \begin{array}{c} v_3 \\ \bullet \end{array} \right)$	$3 := \operatorname{Free} \left(\begin{array}{c} v_1 \\ \bullet \end{array} \xrightarrow{f_1} \begin{array}{c} v_2 \\ \bullet \end{array} \xrightarrow{f_2} \begin{array}{c} v_3 \\ \bullet \end{array} \right)$
fong-spivak-seven-sketches EQ0005	33	0.539	$p = g(f(p)) \quad \text{and} \quad q = f(g(q))$	$p = g(f(p)) \quad \text{and} \quad q = f(g(q))$	$p = g(f(p)) \quad \text{and} \quad q = f(g(q))$
fong-spivak-seven-sketches EQ0078 (4.3)	132	0.549	$\Phi: x^{\operatorname{op}} \times y \rightarrow \operatorname{Bool}$	$\Phi: x^{\operatorname{op}} \times y \rightarrow \operatorname{Bool}$	$\Phi: \mathcal{X}^{\operatorname{op}} \times \mathcal{Y} \rightarrow \operatorname{Bool}$
fong-spivak-seven-sketches EQ0091	142	0.550	$\widehat{F}(p, q) := Q(F(p), q) \quad \text{and} \quad \check{F}(q, p) := Q(q, F(p))$	$\widehat{F}(p, q) := Q(F(p), q) \quad \text{and} \quad \check{F}(q, p) := Q(q, F(p))$	$\widehat{F}(p, q) := Q(F(p), q) \quad \text{and} \quad \check{F}(q, p) := Q(q, F(p))$
fong-spivak-seven-sketches EQ0104 (5.17)	165	0.566	$G + G' := ((V \sqcup V'), [\operatorname{in}, \operatorname{in}'], [\operatorname{out}, \operatorname{out}'], (\iota \sqcup \iota'))$	$G + G' := ((V \sqcup V'), [\operatorname{in}, \operatorname{in}'], [\operatorname{out}, \operatorname{out}'], (\iota \sqcup \iota'))$	$G + G' := ((V \sqcup V'), [\operatorname{in}, \operatorname{in}'], [\operatorname{out}, \operatorname{out}'], (\iota \sqcup \iota'))$
fong-spivak-seven-sketches EQ0221	298	0.568	$x \left(x^{\prime}, x \right) \otimes y \left(y, y^{\prime} \right) \leq \Phi(x, y) \multimap \Phi(x', y')$	$x(x', x) \otimes y(y, y') \leq \Phi(x, y) \multimap \Phi(x', y')$	$\mathcal{X}(x', x) \otimes \mathcal{Y}(y, y') \leq \Phi(x, y) \multimap \Phi(x', y')$
fong-spivak-seven-sketches EQ0253	313	0.570	$\left\{ (y, z) \mid \text{there exists } x \in R^m \text{ such that } y = S(g)(x) \text{ and } S(h)(x) = z \right\}$	$\{(y, z) \mid \text{there exists } x \in R^m \text{ such that } y = S(g)(x) \text{ and } S(h)(x) = z\}$	$\{(y, z) \mid \text{there exists } x \in R^m \text{ such that } y = S(g)(x) \text{ and } S(h)(x) = z\}$
fong-spivak-seven-sketches EQ0043	82	0.593	$2 \operatorname{H}_2 \rightarrow 2 \operatorname{Na} \rightarrow (2 \operatorname{NaOH} + \operatorname{H}_2)$	$2\operatorname{H}_2\operatorname{O} \rightarrow (2\operatorname{Na} \rightarrow (2\operatorname{NaOH} + \operatorname{H}_2))$	$2\operatorname{H}_2\operatorname{O} \rightarrow (2\operatorname{Na} \multimap (2\operatorname{NaOH} + \operatorname{H}_2))$
fong-spivak-seven-sketches EQ0179	272	0.598	$h, \quad 1, \quad 2, \quad 3$	$h, \quad 1, \quad 2, \quad 3$	$h, \quad 1, \quad 2, \quad 3$
fong-spivak-seven-sketches EQ0002 (1.15)	21	0.658	$A = \bigcup_{p \in P} A_p \quad \text{and} \quad \text{if } p \neq q \text{ then } A_p \cap A_q = \emptyset$	$A = \bigcup_{p \in P} A_p \quad \text{and} \quad \text{if } p \neq q \text{ then } A_p \cap A_q = \emptyset$	$A = \bigcup_{p \in P} A_p \quad \text{and} \quad \text{if } p \neq q \text{ then } A_p \cap A_q = \emptyset$
fong-spivak-seven-sketches EQ0158 (7.37)	250	0.658	$\operatorname{Sec}_f(U) := \{s: U \rightarrow X \mid (s; f)(u) = u \text{ for all } u \in U\}$	$\operatorname{Sec}_f(U) := \{s: U \rightarrow X \mid (s; f)(u) = u \text{ for all } u \in U\}$	$\operatorname{Sec}_f(U) := \{s: U \rightarrow X \mid (s \circ f)(u) = u \text{ for all } u \in U\}$
fong-spivak-seven-sketches EQ0147 (7.2)	235	0.661	$((A = \operatorname{go}) \Rightarrow \ddot{S} > 1) \wedge ((A = \operatorname{stop}) \Rightarrow \ddot{S} = 0)$	$((A = \operatorname{go}) \Rightarrow \ddot{S} > 1) \wedge ((A = \operatorname{stop}) \Rightarrow \ddot{S} = 0)$	$((A = \operatorname{go}) \Rightarrow \ddot{S} > 1) \wedge ((A = \operatorname{stop}) \Rightarrow \ddot{S} = 0)$
fong-spivak-seven-sketches EQ0191	282	0.677	$\begin{array}{l} t + u \leq (v + w) + u \quad (\text{monotonicity, } t \leq v + w, u \leq u) \\ = v + (w + u) \quad (\text{associativity}) \\ \leq v + (x + z) \quad (\text{monotonicity, } v \leq v, w + u \leq x + v) \\ = (v + x) + z \quad (\text{associativity}) \\ \leq y + z. \quad (\text{monotonicity, } v + x \leq y, z \leq z) \end{array}$	$\begin{array}{l} t + u \leq (v + w) + u \quad (\text{monotonicity, } t \leq v + w, u \leq u) \\ = v + (w + u) \quad (\text{associativity}) \\ \leq v + (x + z) \quad (\text{monotonicity, } v \leq v, w + u \leq x + v) \\ = (v + x) + z \quad (\text{associativity}) \\ \leq y + z. \quad (\text{monotonicity, } v + x \leq y, z \leq z) \end{array}$	$\begin{array}{l} t + u \leq (v + w) + u \quad (\text{monotonicity, } t \leq v + w, u \leq u) \\ = v + (w + u) \quad (\text{associativity}) \\ \leq v + (x + z) \quad (\text{monotonicity, } v \leq v, w + u \leq x + v) \\ = (v + x) + z \quad (\text{associativity}) \\ \leq y + z. \quad (\text{monotonicity, } v + x \leq y, z \leq z) \end{array}$

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fong-spivak-seven-sketches EQ0224	300	0.684	$\begin{aligned} \Phi: \Psi &= \left(M_X^3 * M_\Phi * M_Y^2 \right) * \left(M_Y^2 * M_\Psi * M_Z^3 \right) \\ &= M_X^3 * M_\Phi * M_Y^4 * M_\Psi * M_Z^3 \\ &= \left(M_X^3 * M_\Phi * M_Y^2 \right) * M_\Psi * M_Z^3 \\ &= \Phi * M_\Psi * M_Z^3 \end{aligned}$	$\Phi : \Psi = \left(M_X^3 * M_\Phi * M_Y^2 \right) * \left(M_Y^2 * M_\Psi * M_Z^3 \right) \\ = M_X^3 * M_\Phi * M_Y^4 * M_\Psi * M_Z^3 \\ = \left(M_X^3 * M_\Phi * M_Y^2 \right) * M_\Psi * M_Z^3 \\ = \Phi * M_\Psi * M_Z^3$ $= \begin{pmatrix} 17 & 20 & 20 \\ 11 & 14 & 14 \\ 14 & 17 & 17 \\ 12 & 9 & 15 \end{pmatrix} \begin{pmatrix} \infty & \infty & \infty & \infty \\ \infty & \infty & 0 & \infty \\ 4 & \infty & \infty & 4 \end{pmatrix} \begin{pmatrix} 0 & 2 & 4 & 5 \\ 4 & 0 & 2 & 3 \\ 2 & 4 & 0 & 1 \\ 1 & 3 & 5 & 0 \end{pmatrix}$ $= \begin{pmatrix} 17 & 20 & 20 \\ 11 & 14 & 14 \\ 14 & 17 & 17 \\ 12 & 9 & 15 \end{pmatrix} \begin{pmatrix} \infty & \infty & \infty & \infty \\ 2 & 4 & 0 & 1 \\ 4 & 6 & 8 & 4 \end{pmatrix}$ $= \begin{pmatrix} 22 & 24 & 20 & 21 \\ 16 & 18 & 14 & 15 \\ 19 & 21 & 17 & 18 \\ 11 & 13 & 9 & 10 \end{pmatrix}$	$\Phi \circ \Psi = (M_X^3 * M_\Phi * M_Y^2) * (M_Y^2 * M_\Psi * M_Z^3) \\ = M_X^3 * M_\Phi * M_Y^4 * M_\Psi * M_Z^3 \\ = (M_X^3 * M_\Phi * M_Y^2) * M_\Psi * M_Z^3 \\ = \Phi * M_\Psi * M_Z^3$
fong-spivak-seven-sketches EQ0120 (5.81)	187	0.690	$\mathbf{SFG}_R^+ := \text{Free} \left(G_R \sqcup G_R^{\text{op}} \right).$	$\mathbf{SFG}_R^+ := \text{Free} \left(G_R \sqcup G_R^{\text{op}} \right).$	$\mathbf{SFG}_R^+ := \text{Free} \left(G_R \sqcup G_R^{\text{op}} \right).$
fong-spivak-seven-sketches EQ0081 (♢)	133	0.690	$x \left(x', x \right) \otimes \Phi(x, y) \otimes y(y, y') \leq \Phi(x', y').$	$x \left(x', x \right) \otimes \Phi(x, y) \otimes y(y, y') \leq \Phi(x', y').$	$\mathcal{X}(x', x) \otimes \Phi(x, y) \otimes \mathcal{Y}(y, y') \leq \Phi(x', y').$
fong-spivak-seven-sketches EQ0140 (6.81)	220	0.725	$\begin{aligned} \psi_{V,V'}: \text{Circ}(V) \times \text{Circ}(V') &\longrightarrow \text{Circ}(V + V'); \\ ((V, A, s, t, \ell), (V', A', s', t', \ell')) &\longmapsto (V + V', A + A', s + s', t + t', [\ell, \ell']). \end{aligned}$	$\psi_{V,V'}: \text{Circ}(V) \times \text{Circ}(V') \longrightarrow \text{Circ}(V + V'); \\ ((V, A, s, t, \ell), (V', A', s', t', \ell')) \longmapsto (V + V', A + A', s + s', t + t', [\ell, \ell']).$	$\psi_{V,V'}: \text{Circ}(V) \times \text{Circ}(V') \longrightarrow \text{Circ}(V + V'); \\ ((V, A, s, t, \ell), (V', A', s', t', \ell')) \longmapsto (V + V', A + A', s + s', t + t', [\ell, \ell']).$
fong-spivak-seven-sketches EQ0222	298	0.731	$\mathcal{X}(x', x) \otimes \Phi(x, y) \otimes y(y, y') \leq \Phi(x', y').$	$\mathcal{X}(x', x) \otimes \Phi(x, y) \otimes y(y, y') \leq \Phi(x', y').$	$\mathcal{X}(x', x) \otimes \Phi(x, y) \otimes \mathcal{Y}(y, y') \leq \Phi(x', y').$

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fong-spivak-seven-sketches-EQ0132	212	0.742	$\begin{array}{rrrr} \operatorname{in}(\mu) := 2, & \operatorname{in}(\eta) := 0, & \operatorname{in}(\delta) := 1, & \operatorname{in}(\epsilon) := 1, \\ \operatorname{out}(\mu) := 1, & \operatorname{out}(\eta) := 1, & \operatorname{out}(\delta) := 2, & \operatorname{out}(\epsilon) := 0. \end{array}$	$\begin{array}{rrrr} \operatorname{in}(\mu) := 2, & \operatorname{in}(\eta) := 0, & \operatorname{in}(\delta) := 1, & \operatorname{in}(\epsilon) := 1, \\ \operatorname{out}(\mu) := 1, & \operatorname{out}(\eta) := 1, & \operatorname{out}(\delta) := 2, & \operatorname{out}(\epsilon) := 0. \end{array}$	$\begin{array}{rrrr} \operatorname{in}(\mu) := 2, & \operatorname{in}(\eta) := 0, & \operatorname{in}(\delta) := 1, & \operatorname{in}(\epsilon) := 1, \\ \operatorname{out}(\mu) := 1, & \operatorname{out}(\eta) := 1, & \operatorname{out}(\delta) := 2, & \operatorname{out}(\epsilon) := 0. \end{array}$
fong-spivak-seven-sketches-EQ0097	155	0.755	$\backslash[\alpha: \beta := \iota_{A \sqcup C}^* ((\iota_{A \sqcup B})!(\alpha) \vee (\iota_{B \sqcup C})!(\beta)).$	$\alpha : \beta := \iota_{A \sqcup C}^* ((\iota_{A \sqcup B})!(\alpha) \vee (\iota_{B \sqcup C})!(\beta)).$	$\alpha \circ \beta := \iota_{A \sqcup C}^* ((\iota_{A \sqcup B})!(\alpha) \vee (\iota_{B \sqcup C})!(\beta)).$
fong-spivak-seven-sketches-EQ0230	304	0.759	$\begin{aligned} & \bigvee_{a,b,c,d,e \in X} \alpha^{-1}(x,(a,1)) \otimes \mathrm{U}_X(a,b) \otimes \eta_X(1,c,d) \otimes \epsilon_X(b,c,1) \otimes \mathrm{U}_X(d,e) \otimes \alpha((1,e),y) \\ & = \bigvee_{a,b,c,d,e \in X} \alpha^{-1}(x,(a,1)) \otimes \mathrm{U}_X(a,b) \otimes \eta_X(1,c,d) \otimes \epsilon_X(b,c,1) \otimes \mathrm{U}_X(d,e) \otimes \alpha((1,e),y) \\ & = \bigvee_{a,b,c,d,e \in X} X(x,a) \otimes X(a,b) \otimes X(c,d) \otimes X(b,c) \otimes X(d,e) \otimes X(e,y) \\ & = X(x,y) \end{aligned}$	$\begin{aligned} & \bigvee_{a,b,c,d,e \in X} \alpha^{-1}(x,(a,1)) \otimes \mathrm{U}_X(a,b) \otimes \eta_X(1,c,d) \otimes \epsilon_X(b,c,1) \otimes \mathrm{U}_X(d,e) \otimes \alpha((1,e),y) \\ & = \bigvee_{a,b,c,d,e \in X} X(x,a) \otimes X(a,b) \otimes X(c,d) \otimes X(b,c) \otimes X(d,e) \otimes X(e,y) \\ & = X(x,y) \end{aligned}$	$\begin{aligned} & \bigvee_{a,b,c,d,e \in X} \alpha^{-1}(x,(a,1)) \otimes \mathrm{U}_X(a,b) \otimes \eta_X(1,c,d) \otimes \epsilon_X(b,c,1) \otimes \mathrm{U}_X(d,e) \otimes \alpha((1,e),y) \\ & = \bigvee_{a,b,c,d,e \in X} X(x,a) \otimes X(a,b) \otimes X(c,d) \otimes X(b,c) \otimes X(d,e) \otimes X(e,y) \\ & = X(x,y) \end{aligned}$
fong-spivak-seven-sketches-EQ0242	310	0.775	$c_n := -c : (1 + -c) : (1 + 1 + -c) : \cdots : (1 + \cdots + 1 + -c) : 1 \rightarrow n$	$c_n := -c : (1 + -c) : (1 + 1 + -c) : \cdots : (1 + \cdots + 1 + -c) : 1 \rightarrow n$	$c_n := -c : (1 + -c) : (1 + 1 + -c) : \cdots : (1 + \cdots + 1 + -c) : 1 \rightarrow n$
fong-spivak-seven-sketches-EQ0156	248	0.780	$\mathbf{Op}_1 := \{\emptyset, \{1\}, X\} \quad \text{and} \quad \mathbf{Op}_2 := \{\emptyset, \{2\}, X\}$	$\mathbf{Op}_1 := \{\emptyset, \{1\}, X\} \quad \text{and} \quad \mathbf{Op}_2 := \{\emptyset, \{2\}, X\}$	$\mathbf{Op}_1 := \{\emptyset, \{1\}, X\} \quad \text{and} \quad \mathbf{Op}_2 := \{\emptyset, \{2\}, X\}$
fong-spivak-seven-sketches-EQ0219	298	0.795	$\begin{aligned} & \Lambda(\text{monoid}, \text{nothing}) = \Lambda(\text{monoid}, \text{this book}) \\ & = \Lambda(\text{preorder}, \text{this book}) = \Lambda(\text{category}, \text{this book}) = \text{true} \\ & \Lambda(\text{preorder}, \text{nothing}) = \Lambda(\text{category}, \text{nothing}) = \text{false}. \end{aligned}$	$\begin{aligned} & \Lambda(\text{monoid}, \text{nothing}) = \Lambda(\text{monoid}, \text{this book}) \\ & = \Lambda(\text{preorder}, \text{this book}) = \Lambda(\text{category}, \text{this book}) = \text{true} \\ & \Lambda(\text{preorder}, \text{nothing}) = \Lambda(\text{category}, \text{nothing}) = \text{false}. \end{aligned}$	$\begin{aligned} & \Lambda(\text{monoid}, \text{nothing}) = \Lambda(\text{monoid}, \text{this book}) \\ & = \Lambda(\text{preorder}, \text{this book}) = \Lambda(\text{category}, \text{this book}) = \text{true} \\ & \Lambda(\text{preorder}, \text{nothing}) = \Lambda(\text{category}, \text{nothing}) = \text{false}. \end{aligned}$
fong-spivak-seven-sketches-EQ0077	124	0.797	$\begin{aligned} & \lim_{\mathcal{J}} D := \{(d_1, \dots, d_n) \mid d_i \in D(v_i) \text{ for all } 1 \leq i \leq n \text{ and} \\ & \text{for all } a : v_i \rightarrow v_j \in A, \text{ we have } D(a)(d_i) = d_j\}. \end{aligned}$	$\lim_{\mathcal{J}} D := \{(d_1, \dots, d_n) \mid d_i \in D(v_i) \text{ for all } 1 \leq i \leq n \text{ and} \\ \text{for all } a : v_i \rightarrow v_j \in A, \text{ we have } D(a)(d_i) = d_j\}.$	$\lim_{\mathcal{J}} D := \{(d_1, \dots, d_n) \mid d_i \in D(v_i) \text{ for all } 1 \leq i \leq n \text{ and} \\ \text{for all } a : v_i \rightarrow v_j \in A, \text{ we have } D(a)(d_i) = d_j\}.$
fong-spivak-seven-sketches-EQ0131	211	0.808	$m \text{ legs } \lesssim n \text{ legs}$	$m \text{ legs } \lesssim n \text{ legs}$	$m \text{ legs } \left\{ \begin{array}{c} \text{diagram} \end{array} \right\} n \text{ legs}$

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fong-spivak-seven-sketches-1 EQ0193	286	0.825	$\begin{aligned} (x \times y) \left(\left(x_1, y_1 \right), \left(x_2, y_2 \right) \right) \otimes (x \times y) \left(\left(x_2, y_2 \right), \left(x_3, y_3 \right) \right) \\ = x \left(x_1, x_2 \right) \otimes y \left(y_1, y_2 \right) \otimes x \left(x_2, x_3 \right) \otimes y \left(y_2, y_3 \right) \\ = x \left(x_1, x_2 \right) \otimes x \left(x_2, x_3 \right) \otimes y \left(y_1, y_2 \right) \otimes y \left(y_2, y_3 \right) \\ \leq x \left(x_1, x_3 \right) \otimes y \left(y_1, y_3 \right) \\ = (x \times y) \left(\left(x_1, y_1 \right), \left(x_3, y_3 \right) \right). \end{aligned}$	$(x \times y) \left(\left(x_1, y_1 \right), \left(x_2, y_2 \right) \right) \otimes (x \times y) \left(\left(x_2, y_2 \right), \left(x_3, y_3 \right) \right)$ $= \mathcal{X}(x_1, x_2) \otimes \mathcal{Y}(y_1, y_2) \otimes \mathcal{X}(x_2, x_3) \otimes \mathcal{Y}(y_2, y_3)$ $= \mathcal{X}(x_1, x_2) \otimes \mathcal{X}(x_2, x_3) \otimes \mathcal{Y}(y_1, y_2) \otimes \mathcal{Y}(y_2, y_3)$ $\leq \mathcal{X}(x_1, x_3) \otimes \mathcal{Y}(y_1, y_3)$ $= (\mathcal{X} \times \mathcal{Y}) \left(\left(x_1, y_1 \right), \left(x_3, y_3 \right) \right).$	
fong-spivak-seven-sketches-1 EQ0172	264	0.827	$\mathbb{I} \mid \mathbb{R} := \{ [d, u] \mid d \leq u \}.$	$\mathbb{I} \mathbb{R} := \{ [d, u] \subseteq \mathbb{R} \mid d \leq u \}.$	$\mathbb{I} \mathbb{R} := \{ [d, u] \subseteq \mathbb{R} \mid d \leq u \}.$
fong-spivak-seven-sketches-1 EQ0093	143	0.828	$\operatorname{Col}(\Phi)(a, b) := \begin{cases} x(a, b) & \text{if } a, b \in X \\ \Phi(a, b) & \text{if } a \in X, b \in Y \\ \emptyset & \text{if } a \in Y, b \in X \\ y(a, b) & \text{if } a, b \in Y \end{cases}$	$\operatorname{Col}(\Phi)(a, b) := \begin{cases} x(a, b) & \text{if } a, b \in X \\ \Phi(a, b) & \text{if } a \in X, b \in Y \\ \emptyset & \text{if } a \in Y, b \in X \\ y(a, b) & \text{if } a, b \in Y \end{cases}$	$\operatorname{Col}(\Phi)(a, b) := \begin{cases} \mathcal{X}(a, b) & \text{if } a, b \in \mathcal{X} \\ \Phi(a, b) & \text{if } a \in \mathcal{X}, b \in \mathcal{Y} \\ \emptyset & \text{if } a \in \mathcal{Y}, b \in \mathcal{X} \\ \mathcal{Y}(a, b) & \text{if } a, b \in \mathcal{Y} \end{cases}$
fong-spivak-seven-sketches-1 EQ0022	58	0.830	$\frac{\frac{x}{y}}{z} = \frac{\frac{x}{y}}{z}$	$\frac{\frac{x}{y}}{z} = \frac{\frac{x}{y}}{z}$	$\frac{\frac{x}{y}}{z} = \frac{\frac{x}{y}}{z}$
fong-spivak-seven-sketches-1 EQ0146	229	0.839	$\operatorname{Circ}(\varphi) : \operatorname{Circ}(2) \times \operatorname{Circ}(2) \times \operatorname{Circ}(2) \rightarrow \operatorname{Circ}(0).$	$\operatorname{Circ}(\varphi) : \operatorname{Circ}(2) \times \operatorname{Circ}(2) \times \operatorname{Circ}(2) \rightarrow \operatorname{Circ}(0).$	$\operatorname{Circ}(\varphi) : \operatorname{Circ}(2) \times \operatorname{Circ}(2) \times \operatorname{Circ}(2) \rightarrow \operatorname{Circ}(0).$
fong-spivak-seven-sketches-1 EQ0086 (4.25)	140	0.842	$\operatorname{U}_x(x, y) := \mathcal{X}(x, y).$	$\operatorname{U}_x(x, y) := \mathcal{X}(x, y).$	$\operatorname{U}_x(x, y) := \mathcal{X}(x, y).$
fong-spivak-seven-sketches-1 EQ0136	219	0.843	$C := \{ \text{light}, \text{switch}, \text{battery} \} \sqcup \{ x\Omega \mid x \in \mathbb{R}^+ \}.$	$C := \{ \text{light}, \text{switch}, \text{battery} \} \sqcup \{ x\Omega \mid x \in \mathbb{R}^+ \}.$	$C := \{ \text{light}, \text{switch}, \text{battery} \} \sqcup \{ x\Omega \mid x \in \mathbb{R}^+ \}.$
fong-spivak-seven-sketches-1 EQ0125	197	0.843	$\mathcal{C} := \stackrel{A}{\bullet} \stackrel{f}{\rightrightarrows} \stackrel{B}{\bullet}$	$\mathcal{C} := \stackrel{A}{\bullet} \stackrel{f}{\rightrightarrows} \stackrel{B}{\bullet}$	$\mathcal{C} := \stackrel{A}{\bullet} \stackrel{f}{\rightrightarrows} \stackrel{B}{\bullet}$
fong-spivak-seven-sketches-1 EQ0003	24	0.849	$S := \begin{array}{ ccc } \hline 11 & 12 & 13 \\ \bullet & & \\ 21 & 22 & 23 \\ \bullet & \bullet & \bullet \\ \hline \end{array}$	$S := \begin{array}{ ccc } \hline 11 & 12 & 13 \\ \bullet & & \\ 21 & 22 & 23 \\ \bullet & \bullet & \bullet \\ \hline \end{array}$	$S := \begin{array}{ ccc } \hline 11 & 12 & 13 \\ \bullet & & \\ 21 & 22 & 23 \\ \bullet & \bullet & \bullet \\ \hline \end{array}$
fong-spivak-seven-sketches-1 EQ0121	188	0.868	$\operatorname{B}(g \circ (h^{\operatorname{op}})) = \{ (x, y) \mid S(g)(x) = S(h)(y) \}.$	$\operatorname{B}(g \circ (h^{\operatorname{op}})) = \{ (x, y) \mid S(g)(x) = S(h)(y) \}.$	$\operatorname{B}(g \circ (h^{\operatorname{op}})) = \{ (x, y) \mid S(g)(x) = S(h)(y) \}.$

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fong-spivak-seven-sketches-EQ0107	174	0.873	$\begin{array}{r} \dot{x}+3\ddot{y}-2z=0 \\ \dot{z}=0 \end{array}$	$\begin{array}{l} \dot{x}+3\ddot{y}-2z=0 \\ \ddot{y}+5\dot{z}=0 \end{array}$	$\begin{array}{l} \dot{x}+3\ddot{y}-2z=0 \\ \ddot{y}+5\dot{z}=0 \end{array}$
fong-spivak-seven-sketches-EQ0010 (1.108)	42	0.877	$p \leq g(f(p)) \quad \text{and} \quad f(g(q)) \leq q$	$p \leq g(f(p)) \quad \text{and} \quad f(g(q)) \leq q.$	$p \leq g(f(p)) \quad \text{and} \quad f(g(q)) \leq q.$
fong-spivak-seven-sketches-EQ0079 (4.5)	132	0.881	$b \wedge c \leq d \quad \text{iff} \quad b \leq (c \Rightarrow d).$	$b \wedge c \leq d \quad \text{iff} \quad b \leq (c \Rightarrow d).$	$b \wedge c \leq d \quad \text{iff} \quad b \leq (c \Rightarrow d).$
fong-spivak-seven-sketches-EQ0217	296	0.890	$\begin{aligned} \lim_{\mathfrak{J}} \{D\} &:= \left\{ \left(d_1, \dots, d_n \right) \mid d_i \in D(v_i) \text{ for all } 1 \leq i \leq n \text{ and} \right. \\ &\quad \left. \text{for all } a : v_i \rightarrow v_j \in A, \text{ we have } D(a)(d_i) = d_j \right\}. \end{aligned}$	$\lim_{\mathfrak{J}} D := \{(d_1, \dots, d_n) \mid d_i \in D(v_i) \text{ for all } 1 \leq i \leq n \text{ and for all } a : v_i \rightarrow v_j \in A, \text{ we have } D(a)(d_i) = d_j\}.$	$\lim_{\mathfrak{J}} D := \{(d_1, \dots, d_n) \mid d_i \in D(v_i) \text{ for all } 1 \leq i \leq n \text{ and for all } a : v_i \rightarrow v_j \in A, \text{ we have } D(a)(d_i) = d_j\}.$
fong-spivak-seven-sketches-EQ0153 (7.18)	242	0.894	$X = \{(\text{true}, \text{true}), (\text{true}, \text{false}), (\text{false}, \text{true}), (\text{false}, \text{false})\} \subseteq \mathbb{B} \times \mathbb{B}$	$X = \{(\text{true}, \text{true}), (\text{true}, \text{false}), (\text{false}, \text{true})\} \subseteq \mathbb{B} \times \mathbb{B}.$	$X = \{(\text{true}, \text{true}), (\text{true}, \text{false}), (\text{false}, \text{true})\} \subseteq \mathbb{B} \times \mathbb{B}.$
fong-spivak-seven-sketches-EQ0105	167	0.897	$a \subset \bullet \supset b$	$a \subset \bullet \supset b$	$a \subset \bullet \supset b$
fong-spivak-seven-sketches-EQ0143 (6.95)	227	0.902	$\left(g \circ \text{circ}\right)_i: X_1 \times \cdots \times X_{i-1} \times W_1 \times \cdots \times W_m \times X_{i+1} \times \cdots \times X_n \longrightarrow Y$	$(g \circ_i f): X_1 \times \cdots \times X_{i-1} \times W_1 \times \cdots \times W_m \times X_{i+1} \times \cdots \times X_n \longrightarrow Y$	$(g \circ_i f): X_1 \times \cdots \times X_{i-1} \times W_1 \times \cdots \times W_m \times X_{i+1} \times \cdots \times X_n \longrightarrow Y$
fong-spivak-seven-sketches-EQ0129	205	0.913	$\left\{ (v, d) \mid d \in D(v), \text{ where } v = v_1 \text{ or } v = v_2 \right\}.$	$\{(v, d) \mid d \in D(v), \text{ where } v = v_1 \text{ or } v = v_2\}.$	$\{(v, d) \mid d \in D(v), \text{ where } v = v_1 \text{ or } v = v_2\}.$
fong-spivak-seven-sketches-EQ0164	257	0.917	$\stackrel{A}{\longrightarrow} \bullet \stackrel{B}{\longrightarrow} \bullet \stackrel{C}{\longrightarrow} \bullet$	$\stackrel{A}{\longrightarrow} \bullet \stackrel{B}{\longrightarrow} \bullet \stackrel{C}{\longrightarrow} \bullet$	$\stackrel{A}{\longrightarrow} \bullet \stackrel{B}{\longrightarrow} \bullet \stackrel{C}{\longrightarrow} \bullet$
fong-spivak-seven-sketches-EQ0256	315	0.919	$\begin{aligned} f(m) *_{\mathbb{R}} f(n) &= \underbrace{(1_{\mathbb{R}} + \cdots + 1_{\mathbb{R}})}_{m \text{ summands}} * \underbrace{(1_{\mathbb{R}} + \cdots + 1_{\mathbb{R}})}_{n \text{ summands}} \\ &= 1_{\mathbb{R}} * \underbrace{(1_{\mathbb{R}} + \cdots + 1_{\mathbb{R}})}_{n \text{ summands}} + \cdots + 1_{\mathbb{R}} * \underbrace{(1_{\mathbb{R}} + \cdots + 1_{\mathbb{R}})}_{n \text{ summands}} \\ &= \underbrace{1_{\mathbb{R}} + \cdots + 1_{\mathbb{R}}}_{mn \text{ summands}} = f(m * n). \end{aligned}$	$\begin{aligned} f(m) *_{\mathbb{R}} f(n) &= \underbrace{(1_{\mathbb{R}} + \cdots + 1_{\mathbb{R}})}_{m \text{ summands}} * \underbrace{(1_{\mathbb{R}} + \cdots + 1_{\mathbb{R}})}_{n \text{ summands}} \\ &= 1_{\mathbb{R}} * \underbrace{(1_{\mathbb{R}} + \cdots + 1_{\mathbb{R}})}_{n \text{ summands}} + \cdots + 1_{\mathbb{R}} * \underbrace{(1_{\mathbb{R}} + \cdots + 1_{\mathbb{R}})}_{n \text{ summands}} \\ &= \underbrace{1_{\mathbb{R}} + \cdots + 1_{\mathbb{R}}}_{mn \text{ summands}} = f(m * n). \end{aligned}$	$\begin{aligned} f(m) *_{\mathbb{R}} f(n) &= \underbrace{(1_{\mathbb{R}} + \cdots + 1_{\mathbb{R}})}_{m \text{ summands}} *_{\mathbb{R}} \underbrace{(1_{\mathbb{R}} + \cdots + 1_{\mathbb{R}})}_{n \text{ summands}} \\ &= 1_{\mathbb{R}} * \underbrace{(1_{\mathbb{R}} + \cdots + 1_{\mathbb{R}})}_{n \text{ summands}} + \cdots + 1_{\mathbb{R}} * \underbrace{(1_{\mathbb{R}} + \cdots + 1_{\mathbb{R}})}_{n \text{ summands}} \\ &= \underbrace{1_{\mathbb{R}} + \cdots + 1_{\mathbb{R}}}_{mn \text{ summands}} = f(m * n). \end{aligned}$
fong-spivak-seven-sketches-EQ0096	149	0.919	$(f \times g) \circ (f' \times g') = (f \circ f') \times (g \circ g')$	$(f \times g); (f' \times g') = (f; g) \times (f'; g').$	$(f \times g) \circ (f' \times g') = (f \circ f') \times (g \circ g').$

Identifier	Page	Conf.	LaTeX source	Rendered	Image
fong-spivak-seven-sketches_3EQ0259	321	■0.923	$\begin{aligned} \varphi_{S',T'}(\varphi_{S,T}(A \times B)) &= \varphi_{S',T'}(\{f(a) \mid a \in A\} \times \{g(b) \mid b \in B\}) \\ &= \{(f(a),g(b)) \mid a \in A, b \in B\} \\ &= \operatorname{im}_{f \times g}(A \times B) \\ &= \operatorname{im}_{f \times g}(\varphi_{S,T}(A, B)). \end{aligned}$	$\varphi_{S',T'}((\operatorname{im}_f \times \operatorname{im}_g)(A \times B)) = \varphi_{S',T'}(\{f(a) \mid a \in A\} \times \{g(b) \mid b \in B\})$ $= \{(f(a),g(b)) \mid a \in A, b \in B\}$ $= \operatorname{im}_{f \times g}(A \times B)$ $= \operatorname{im}_{f \times g}(\varphi_{S,T}(A, B)).$	$\varphi_{S',T'}\left((\operatorname{im}_f \times \operatorname{im}_g)(A \times B)\right) = \varphi_{S',T'}(\{f(a) \mid a \in A\} \times \{g(b) \mid b \in B\})$ $= \{(f(a),g(b)) \mid a \in A, b \in B\}$ $= \operatorname{im}_{f \times g}(A \times B)$ $= \operatorname{im}_{f \times g}(\varphi_{S,T}(A, B)).$
fong-spivak-seven-sketches_1EQ0087	140	■0.928	$\mathrm{U}_{\mathcal{P}}; \Phi = \Phi = \Phi; \mathrm{U}_{\mathcal{Q}}$	$\mathrm{U}_{\mathcal{P}}; \Phi = \Phi = \Phi; \mathrm{U}_{\mathcal{Q}}$	$\mathrm{U}_{\mathcal{P}} \circ \Phi = \Phi = \Phi \circ \mathrm{U}_{\mathcal{Q}}$
fong-spivak-seven-sketches_1EQ0199	288	■0.928	$\begin{aligned} ((M * N) * P)(w, z) &= \bigvee_{y \in Y} \left(\bigvee_{x \in X} M(w, x) \otimes N(x, y) \right) \otimes P(y, z) \\ &\cong \bigvee_{y \in Y, x \in X} M(w, x) \otimes N(x, y) \otimes P(y, z) \\ &\cong \bigvee_{x \in X} M(w, x) \otimes \left(\bigvee_{y \in Y} N(x, y) \otimes P(y, z) \right) \\ &= (M * (N * P))(w, z). \end{aligned}$	$((M * N) * P)(w, z) = \bigvee_{y \in Y} \left(\bigvee_{x \in X} M(w, x) \otimes N(x, y) \right) \otimes P(y, z)$ $\cong \bigvee_{y \in Y, x \in X} M(w, x) \otimes N(x, y) \otimes P(y, z)$ $\cong \bigvee_{x \in X} M(w, x) \otimes \left(\bigvee_{y \in Y} N(x, y) \otimes P(y, z) \right)$ $= (M * (N * P))(w, z).$	$((M * N) * P)(w, z) = \bigvee_{y \in Y} \left(\bigvee_{x \in X} M(w, x) \otimes N(x, y) \right) \otimes P(y, z)$ $\cong \bigvee_{y \in Y, x \in X} M(w, x) \otimes N(x, y) \otimes P(y, z)$ $\cong \bigvee_{x \in X} M(w, x) \otimes \left(\bigvee_{y \in Y} N(x, y) \otimes P(y, z) \right)$ $= (M * (N * P))(w, z).$
fong-spivak-seven-sketches_3EQ0251	313	■0.930	$\mathrm{B}(g; (h^{\mathrm{op}})) = \{(x, y) \mid S(g)(x) = S(h)(y)\}.$	$\mathrm{B}(g; (h^{\mathrm{op}})) = \{(x, y) \mid S(g)(x) = S(h)(y)\}.$	$\mathrm{B}(g \circ (h^{\mathrm{op}})) = \{(x, y) \mid S(g)(x) = S(h)(y)\}.$
fong-spivak-seven-sketches_1EQ0034	73	■0.932	$d_L(U, V) := \sup_{u \in U} \inf_{v \in V} d(u, v),$	$d_L(U, V) := \sup_{u \in U} \inf_{v \in V} d(u, v),$	$d_L(U, V) := \sup_{u \in U} \inf_{v \in V} d(u, v),$
fong-spivak-seven-sketches_1EQ0128	205	■0.939	$\operatorname{colim}_{\mathcal{J}} D := \{(v, d) \mid v \in V \text{ and } d \in D(v)\} / \sim$	$\operatorname{colim}_{\mathcal{J}} D := \{(v, d) \mid v \in V \text{ and } d \in D(v)\} / \sim$	$\operatorname{colim}_{\mathcal{J}} D := \{(v, d) \mid v \in V \text{ and } d \in D(v)\} / \sim$
fong-spivak-seven-sketches_1EQ0053 (3.8)	94	■0.941	$\mathbf{2} := \operatorname{Free} \left(\begin{array}{ ccc } \hline v_1 & f_1 & v_2 \\ \hline \bullet & & \bullet \\ \hline \end{array} \right)$	$\mathbf{2} := \operatorname{Free} \left(\begin{array}{ ccc } \hline v_1 & f_1 & v_2 \\ \hline \bullet & & \bullet \\ \hline \end{array} \right)$	$\mathbf{2} := \operatorname{Free} \left(\begin{array}{ ccc } \hline v_1 & f_1 & v_2 \\ \hline \bullet & \longrightarrow & \bullet \\ \hline \end{array} \right)$
fong-spivak-seven-sketches_1EQ0075	119	■0.943	$\frac{1}{\operatorname{Em}_6}$	<i>(not rendered)</i>	$\frac{\mathbf{1}}{\operatorname{Em}_6}$
fong-spivak-seven-sketches_1EQ0026	60	■0.948	$\mathrm{H}_2\mathrm{O}, \quad \mathrm{NaCl}, \quad 2\mathrm{NaOH}, \quad \mathrm{CH}_4 + 3\mathrm{O}_2$	$\mathrm{H}_2\mathrm{O}, \quad \mathrm{NaCl}, \quad 2\mathrm{NaOH}, \quad \mathrm{CH}_4 + 3\mathrm{O}_2$	$\mathrm{H}_2\mathrm{O}, \quad \mathrm{NaCl}, \quad 2\mathrm{NaOH}, \quad \mathrm{CH}_4 + 3\mathrm{O}_2$
fong-spivak-seven-sketches_1EQ0111	176	■0.950	$\left(\begin{array}{cc} A & 0 \\ 0 & B \end{array} \right)$	$\left(\begin{array}{cc} A & 0 \\ 0 & B \end{array} \right)$	$\left(\begin{array}{cc} A & 0 \\ 0 & B \end{array} \right)$
fong-spivak-seven-sketches_1EQ0159 (7.43)	252	■0.951	$\operatorname{Sec}_f(U) := \{g: U \rightarrow X \mid g \text{ is continuous and } (g \circ f)(u) = u \text{ for all } u \in U\},$	$\operatorname{Sec}_f(U) := \{g: U \rightarrow X \mid g \text{ is continuous and } (g \circ f)(u) = u \text{ for all } u \in U\},$	$\operatorname{Sec}_f(U) := \{g: U \rightarrow X \mid g \text{ is continuous and } (g \circ f)(u) = u \text{ for all } u \in U\},$
fong-spivak-seven-sketches_3EQ0007	38	■0.952	$f(a) \vee f(b) \not\equiv f(a \vee b) .$	$f(a) \vee f(b) \not\equiv f(a \vee b).$	$f(a) \vee f(b) \not\equiv f(a \vee b).$

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fong-spivak-seven-sketchesEQ0030 (2.25)	62	■ 0.954	$\mathrm{H}_2 \mathrm{O} + \mathrm{NaCl} \rightarrow^? \mathrm{H}_2 \mathrm{O}$	$\mathrm{H}_2\mathrm{O} + \mathrm{NaCl} \rightarrow^? \mathrm{H}_2\mathrm{O}$	$\mathrm{H}_2\mathrm{O} + \mathrm{NaCl} \rightarrow^? \mathrm{H}_2\mathrm{O}$
fong-spivak-seven-sketchesEQ0194	287	■ 0.955	$A = (A \cap \bar{B}) \cup (A \cap B) \subseteq \bar{B} \cup C$.	$A = (A \cap \bar{B}) \cup (A \cap B) \subseteq \bar{B} \cup C.$	$A = (A \cap \bar{B}) \cup (A \cap B) \subseteq \bar{B} \cup C.$
fong-spivak-seven-sketchesEQ0171 (7.73)	263	■ 0.958	$\forall (t : T) . \exists (s : S) . f(s) = t$	$\forall (t : T) . \exists (s : S) . f(s) = t$	$\forall (t : T) . \exists (s : S) . f(s) = t$
fong-spivak-seven-sketchesEQ0040 (2.80)	81	■ 0.960	$(a \otimes v) \leq w \quad \text{iff} \quad a \leq (v \multimap w)$.	$(a \otimes v) \leq w \quad \text{iff} \quad a \leq (v \multimap w).$	$(a \otimes v) \leq w \quad \text{iff} \quad a \leq (v \multimap w).$
fong-spivak-seven-sketchesEQ0262	328	■ 0.963	$\bigcup_{i \in I} A_i = \bigcup_{i \in I} \left(B_i \cap Y \right) \cap Y \in \mathbf{Op}_{? \cap Y}.$	$\bigcup_{i \in I} A_i = \bigcup_{i \in I} (B_i \cap Y) = \left(\bigcup_{i \in I} B_i \right) \cap Y \in \mathbf{Op}_{? \cap Y}.$	$\bigcup_{i \in I} A_i = \bigcup_{i \in I} (B_i \cap Y) = \left(\bigcup_{i \in I} B_i \right) \cap Y \in \mathbf{Op}_{? \cap Y}.$
fong-spivak-seven-sketchesEQ0092 (4.40)	143	■ 0.967	$\mathcal{P}(p, G(q)) \cong \mathcal{Q}(F(p), q)$	$\mathcal{P}(p, G(q)) \cong \mathcal{Q}(F(p), q)$	$\mathcal{P}(p, G(q)) \cong \mathcal{Q}(F(p), q)$
fong-spivak-seven-sketchesEQ0218	297	■ 0.967	$\lim_{\mathcal{J}} D := \{(d_1, d_2) \mid d_1 \in A \text{ and } d_2 \in B\}.$	$\lim_{\mathcal{J}} D := \{(d_1, d_2) \mid d_1 \in A \text{ and } d_2 \in B\}.$	$\lim_{\mathcal{J}} D := \{(d_1, d_2) \mid d_1 \in A \text{ and } d_2 \in B\}.$
fong-spivak-seven-sketchesEQ0168	260	■ 0.967	$s : S \mid p(s) \vdash q(s) \quad \text{or} \quad p(s) \vdash_{s:S} q(s).$	$s : S \mid p(s) \vdash q(s) \quad \text{or} \quad p(s) \vdash_{s:S} q(s).$	$s : S \mid p(s) \vdash q(s) \quad \text{or} \quad p(s) \vdash_{s:S} q(s).$
fong-spivak-seven-sketchesEQ0066 (3.69)	114	■ 0.968	$f(p) \leq q \text{ if and only if } p \leq g(q) .$	$f(p) \leq q \text{ if and only if } p \leq g(q).$	$f(p) \leq q \text{ if and only if } p \leq g(q).$
fong-spivak-seven-sketchesEQ0119 (5.78)	187	■ 0.969	$B_1 ; B_2 := \{(x, z) \mid \text{there exists } y \in R^n \text{ such that } (x, y) \in B_1 \text{ and } (y, z) \in B_2\}.$	$B_1; B_2 := \{(x, z) \mid \text{there exists } y \in R^n \text{ such that } (x, y) \in B_1 \text{ and } (y, z) \in B_2\}.$	$B_1 \circ B_2 := \{(x, z) \mid \text{there exists } y \in R^n \text{ such that } (x, y) \in B_1 \text{ and } (y, z) \in B_2\}.$
fong-spivak-seven-sketchesEQ0157	249	■ 0.969	$\left. s_i \right _{U_i \cap U_j} = \left. s_j \right _{U_i \cap U_j}.$	$s_i _{U_i \cap U_j} = s_j _{U_i \cap U_j}.$	$s_i _{U_i \cap U_j} = s_j _{U_i \cap U_j}.$
fong-spivak-seven-sketchesEQ0100	157	■ 0.973	$\eta_x : \mathbf{1} \times \mathcal{X}^{\mathrm{op}} \times \mathcal{X} \rightarrow \mathcal{V};$	$\eta_x : \mathbf{1} \times \mathcal{X}^{\mathrm{op}} \times \mathcal{X} \rightarrow \mathcal{V};$	$\eta_x : \mathbf{1} \times \mathcal{X}^{\mathrm{op}} \times \mathcal{X} \rightarrow \mathcal{V};$
fong-spivak-seven-sketchesEQ0045	85	■ 0.974	$d(U, V) := \sup_{u \in U} \inf_{v \in V} d(u, v) \quad X(U, V) := \bigwedge_{u \in U} \bigvee_{v \in V} x(u, v).$	$d(U, V) := \sup_{u \in U} \inf_{v \in V} d(u, v) \quad X(U, V) := \bigwedge_{u \in U} \bigvee_{v \in V} x(u, v).$	$d(U, V) := \sup_{u \in U} \inf_{v \in V} d(u, v) \quad \mathcal{X}(U, V) := \bigwedge_{u \in U} \bigvee_{v \in V} \mathcal{X}(u, v).$
fong-spivak-seven-sketchesEQ0046	85	■ 0.974	$v \rightarrow w := \bigvee_{\{a \in V \mid a \otimes v \leq w\}} a.$	$v \rightarrow w := \bigvee_{\{a \in V \mid a \otimes v \leq w\}} a.$	$v \multimap w := \bigvee_{\{a \in V \mid a \otimes v \leq w\}} a.$
fong-spivak-seven-sketchesEQ0085	139	■ 0.976	$(\Phi ; \Psi)(p, r) = \bigvee_{q \in Q} (\Phi(p, q) \otimes \Psi(q, r)).$	$(\Phi; \Psi)(p, r) = \bigvee_{q \in Q} (\Phi(p, q) \otimes \Psi(q, r)).$	$(\Phi \circ \Psi)(p, r) = \bigvee_{q \in Q} (\Phi(p, q) \otimes \Psi(q, r)).$
fong-spivak-seven-sketchesEQ0195	287	■ 0.979	$A \cap B \subseteq (\bar{B} \cup C) \cap B = (\bar{B} \cap B) \cup (C \cap B) = C \cap B \subseteq C.$	$A \cap B \subseteq (\bar{B} \cup C) \cap B = (\bar{B} \cap B) \cup (C \cap B) = C \cap B \subseteq C.$	$A \cap B \subseteq (\bar{B} \cup C) \cap B = (\bar{B} \cap B) \cup (C \cap B) = C \cap B \subseteq C.$
fong-spivak-seven-sketchesEQ0173	265	■ 0.979	$\mathbb{R} \cong \{[d, u] \in \mathbb{I}\mathbb{R} \mid d = u\}.$	$\mathbb{R} \cong \{[d, u] \in \mathbb{I}\mathbb{R} \mid d = u\}.$	$\mathbb{R} \cong \{[d, u] \in \mathbb{I}\mathbb{R} \mid d = u\}.$
fong-spivak-seven-sketchesEQ0137	220	■ 0.982	$\text{Circ: } (\mathbf{FinSet}, +) \longrightarrow (\mathbf{Set}, \times)$	$\text{Circ: } (\mathbf{FinSet}, +) \longrightarrow (\mathbf{Set}, \times)$	$\text{Circ: } (\mathbf{FinSet}, +) \longrightarrow (\mathbf{Set}, \times)$
fong-spivak-seven-sketchesEQ0169	260	■ 0.983	$\forall (t : T) . p(s, t) .$	$\forall (t : T) . p(s, t).$	$\forall (t : T) . p(s, t).$
fong-spivak-seven-sketchesEQ0018	46	■ 0.985	$\operatorname{fix}_j := \{p \in P \mid j(p) \cong p\} .$	$\operatorname{fix}_j := \{p \in P \mid j(p) \cong p\}.$	$\operatorname{fix}_j := \{p \in P \mid j(p) \cong p\}.$

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fong-spivak-seven-sketches- EQ0103	165	■ 0.985	$\iota^{\mathrm{\prime \prime}}(x)=\begin{cases}\iota(x) & \text{ if } \iota(x) \in I \\ \iota'(\iota(x)) & \text{ if } \iota(x) \in \underline{n} \\ \iota'(x) & \text{ if } x \in O.\end{cases}$		$\iota''(x)=\begin{cases}\iota(x) & \text{ if } \iota(x) \in I \\ \iota'(\iota(x)) & \text{ if } \iota(x) \in \underline{n} \\ \iota'(x) & \text{ if } x \in O.\end{cases}$
fong-spivak-seven-sketches- EQ0210	292	■ 0.986	$F(a):=a^{\mathrm{\prime}}, \quad G(a):=a^{\mathrm{\prime}}, \quad F(b):=b^{\mathrm{\prime}}, \quad G(b):=b^{\mathrm{\prime}}, \quad F(f):=f_1, \quad \text{ and } G(f):=f_2.$		$F(a) := a', \quad G(a) := a', \quad F(b) := b', \quad G(b) := b', \quad F(f) := f_1, \text{ and } G(f) := f_2.$
fong-spivak-seven-sketches- EQ0117 (5.75)	186	■ 0.988	$\mathrm{B}(g)=\left\{(x, S(g)(x)) \mid x \in R^m\right\} \subseteq R^m \times R^n$	$B(g)=\{(x, S(g)(x)) \mid x \in R^m\} \subseteq R^m \times R^n.$	$B(g)=\{(x, S(g)(x)) \mid x \in R^m\} \subseteq R^m \times R^n.$
fong-spivak-seven-sketches- EQ0031	67	■ 0.989	$\text{Cost} := ([0, \infty], \geq, 0, +),$	$\text{Cost} := ([0, \infty], \geq, 0, +),$	$\text{Cost} := ([0, \infty], \geq, 0, +),$
fong-spivak-seven-sketches- EQ0204	289	■ 0.989	$\stackrel{\bullet}{v_1} \xrightarrow{f_1} \stackrel{\bullet}{v_2} \xrightarrow{f_2} \dots \xrightarrow{f_{n-1}} \stackrel{\bullet}{v_n}$		$\stackrel{\bullet}{v_1} \xrightarrow{f_1} \stackrel{\bullet}{v_2} \xrightarrow{f_2} \dots \xrightarrow{f_{n-1}} \stackrel{\bullet}{v_n}$
fong-spivak-seven-sketches- EQ0213	294	■ 0.989	$G:=\stackrel{1}{\bullet} \xrightarrow{a} \stackrel{2}{\bullet} \xrightarrow{b} \stackrel{3}{\bullet}$	$G := \stackrel{1}{\bullet} \xrightarrow{a} \stackrel{2}{\bullet} \xrightarrow{b} \stackrel{3}{\bullet}$	$G := \left \begin{array}{c} 1 \xrightarrow{a} 2 \xrightarrow{b} 3 \end{array} \right $
fong-spivak-seven-sketches- EQ0250	313	■ 0.990	$\left\{(x, y) \mid \text{there exists } z \in R^n \text{ such that } S(g)(x)=z \text{ and } z=S(h)(y)\right\}.$	$\{(x, y) \mid \text{ there exists } z \in R^n \text{ such that } S(g)(x) = z \text{ and } z = S(h)(y)\}.$	$\{(x, y) \mid \text{ there exists } z \in R^n \text{ such that } S(g)(x) = z \text{ and } z = S(h)(y)\}.$
fong-spivak-seven-sketches- EQ0038	78	■ 0.990	$\mathcal{X}(x_1, x_2) \leq y(F(x_1), F(x_2)),$	$\mathcal{X}(x_1, x_2) \leq y(F(x_1), F(x_2)),$	$\mathcal{X}(x_1, x_2) \leq \mathcal{Y}(F(x_1), F(x_2)),$
fong-spivak-seven-sketches- EQ0102	162	■ 0.991	$R;S:=\{(i,k) \in \underline{m} \times \underline{p} \mid \exists(j \in \underline{n}).(i,j) \in R \text{ and } (j,k) \in S\}.$	$R;S := \{(i, k) \in \underline{m} \times \underline{p} \mid \exists(j \in \underline{n}).(i, j) \in R \text{ and } (j, k) \in S\}.$	$R \circ S := \{(i, k) \in \underline{m} \times \underline{p} \mid \exists(j \in \underline{n}).(i, j) \in R \text{ and } (j, k) \in S\}.$
fong-spivak-seven-sketches- EQ0080 (4.6)	133	■ 0.991	$\Phi: x^{\mathrm{op}} \times y \rightarrow \mathcal{V}.$	$\Phi: x^{\mathrm{op}} \times y \rightarrow \mathcal{V}.$	$\Phi: \mathcal{X}^{\mathrm{op}} \times \mathcal{Y} \rightarrow \mathcal{V}.$
fong-spivak-seven-sketches- EQ0134	216	■ 0.992	$\varphi_{c_1, c_2}: F(c_1) \otimes_{\mathcal{D}} F(c_2) \rightarrow F(c_1 \otimes_{\mathcal{C}} c_2),$	$\varphi_{c_1, c_2}: F(c_1) \otimes_{\mathcal{D}} F(c_2) \rightarrow F(c_1 \otimes_{\mathcal{C}} c_2),$	$\varphi_{c_1, c_2}: F(c_1) \otimes_{\mathcal{D}} F(c_2) \rightarrow F(c_1 \otimes_{\mathcal{C}} c_2),$
fong-spivak-seven-sketches- EQ0148	236	■ 0.992	$\{p \in P \mid \text{likes_cats}(p)\}.$	<i>(not rendered)</i>	$\{p \in P \mid \text{likes_cats}(p)\}.$
fong-spivak-seven-sketches- EQ0144	228	■ 0.992	$(f \circ i \circ g):=f \circ (\mathrm{id}, \ldots, \mathrm{id}, g, \mathrm{id}, \ldots, \mathrm{id})$	$(f \circ ig) := f \circ (\mathrm{id}, \ldots, \mathrm{id}, g, \mathrm{id}, \ldots, \mathrm{id})$	$(f \circ_i g) := f \circ (\mathrm{id}, \ldots, \mathrm{id}, g, \mathrm{id}, \ldots, \mathrm{id})$
fong-spivak-seven-sketches- EQ0123 (5.86)	189	■ 0.993	$\{(0,(x,x)) \mid x \in R\} \subseteq R^0 \times R^2 \quad \text{ and } \quad \{((x,x),0) \mid x \in R\} \subseteq R^2 \times R^0.$	$\{(0,(x,x)) \mid x \in R\} \subseteq R^0 \times R^2 \quad \text{ and } \quad \{((x,x),0) \mid x \in R\} \subseteq R^2 \times R^0.$	$\{(0,(x,x)) \mid x \in R\} \subseteq R^0 \times R^2 \quad \text{ and } \quad \{((x,x),0) \mid x \in R\} \subseteq R^2 \times R^0.$
fong-spivak-seven-sketches- EQ0229	304	■ 0.993	$x \xrightarrow{\alpha^{-1}} x \times \mathbf{1} \xrightarrow{U_x \times \eta_x} x \times x^{\mathrm{op}} \times x \xrightarrow{\epsilon_x \times U_x} \mathbf{1} \times x \xrightarrow{\alpha} x$	$x \xrightarrow{\alpha^{-1}} x \times \mathbf{1} \xrightarrow{U_x \times \eta_x} x \times x^{\mathrm{op}} \times x \xrightarrow{\epsilon_x \times U_x} \mathbf{1} \times x \xrightarrow{\alpha} x$	$\mathcal{X} \xrightarrow{\alpha^{-1}} \mathcal{X} \times \mathbf{1} \xrightarrow{U_{\mathcal{X}} \times \eta_{\mathcal{X}}} \mathcal{X} \times \mathcal{X}^{\mathrm{op}} \times \mathcal{X} \xrightarrow{\epsilon_{\mathcal{X}} \times U_{\mathcal{X}}} \mathbf{1} \times \mathcal{X} \xrightarrow{\alpha} \mathcal{X}$

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fong-spivak-seven-sketches_1 EQ0225	301	■ 0.993	$\begin{aligned} & ((\Phi ; \Psi) ; \Upsilon)(p, s) \ \& \ = \bigvee_{r \in \mathcal{R}} \bigvee_{q \in \mathcal{Q}} \Phi(p, q) \otimes \Psi(q, r) \bigvee_{r \in \mathcal{R}} \bigvee_{q \in \mathcal{Q}} \Phi(p, q) \otimes \Psi(q, r) \otimes \Upsilon(r, s) \\ & \qquad \qquad \qquad = \bigvee_{r \in \mathcal{R}, q \in \mathcal{Q}} \Phi(p, q) \otimes \Psi(q, r) \otimes \Upsilon(r, s) \\ & \qquad \qquad \qquad = \bigvee_{q \in \mathcal{Q}} \Phi(p, q) \otimes \bigvee_{r \in \mathcal{R}} \left(\Psi(q, r) \otimes \Upsilon(r, s) \right) \\ & \qquad \qquad \qquad = (\Phi \circ (\Psi \circ \Upsilon))(p, s) \end{aligned}$	$\begin{aligned} ((\Phi; \Psi); \Upsilon)(p, s) &= \bigvee_{r \in \mathcal{R}} \left(\bigvee_{q \in \mathcal{Q}} \Phi(p, q) \otimes \Psi(q, r) \right) \otimes \Upsilon(r, s) \\ &= \bigvee_{r \in \mathcal{R}, q \in \mathcal{Q}} \Phi(p, q) \otimes \Psi(q, r) \otimes \Upsilon(r, s) \\ &= \bigvee_{q \in \mathcal{Q}} \Phi(p, q) \otimes \bigvee_{r \in \mathcal{R}} (\Psi(q, r) \otimes \Upsilon(r, s)) \\ &= (\Phi; (\Psi; \Upsilon))(p, s) \end{aligned}$	$\begin{aligned} ((\Phi \circ \Psi) \circ \Upsilon)(p, s) &= \bigvee_{r \in \mathcal{R}} \left(\bigvee_{q \in \mathcal{Q}} \Phi(p, q) \otimes \Psi(q, r) \right) \otimes \Upsilon(r, s) \\ &= \bigvee_{r \in \mathcal{R}, q \in \mathcal{Q}} \Phi(p, q) \otimes \Psi(q, r) \otimes \Upsilon(r, s) \\ &= \bigvee_{q \in \mathcal{Q}} \Phi(p, q) \otimes \bigvee_{r \in \mathcal{R}} (\Psi(q, r) \otimes \Upsilon(r, s)) \\ &= (\Phi \circ (\Psi \circ \Upsilon))(p, s) \end{aligned}$
fong-spivak-seven-sketches_1 EQ0089 (4.29)	140	■ 0.994	$\mathcal{P} \left(p, p_1 \right) \otimes \Phi \left(p_1, q \right) = \mathcal{P} \left(p, p_1 \right) \otimes \Phi \left(p_1, q \right) \otimes I \leq \mathcal{P} \left(p, p_1 \right) \otimes \Phi \left(p_1, q \right) \otimes \mathcal{Q} \left(q, q \right) \leq \Phi \left(p, q \right).$	$\mathcal{P}(p, p_1) \otimes \Phi(p_1, q) = \mathcal{P}(p, p_1) \otimes \Phi(p_1, q) \otimes I \leq \mathcal{P}(p, p_1) \otimes \Phi(p_1, q) \otimes \mathcal{Q}(q, q) \leq \Phi(p, q).$	$\mathcal{P}(p, p_1) \otimes \Phi(p_1, q) = \mathcal{P}(p, p_1) \otimes \Phi(p_1, q) \otimes I \leq \mathcal{P}(p, p_1) \otimes \Phi(p_1, q) \otimes \mathcal{Q}(q, q) \leq \Phi(p, q).$
fong-spivak-seven-sketches_1 EQ0165 (7.56)	257	■ 0.995	$(U \wedge V) := U \cap V \quad \text{and} \quad (U \vee V) := U \cup V.$	$(U \wedge V) := U \cap V \quad \text{and} \quad (U \vee V) := U \cup V.$	$(U \wedge V) := U \cap V \quad \text{and} \quad (U \vee V) := U \cup V.$
fong-spivak-seven-sketches_1 EQ0150 (7.14)	241	■ 0.995	$\Omega_{\text{Set}} := \mathbb{B} = \{ \text{true}, \text{false} \}.$	$\Omega_{\text{Set}} := \mathbb{B} = \{ \text{true}, \text{false} \}.$	$\Omega_{\text{Set}} := \mathbb{B} = \{ \text{true}, \text{false} \}.$
fong-spivak-seven-sketches_1 EQ0160 (7.50)	255	■ 0.996	$\Omega(U) := \{ U' \in \mathbf{Op} \mid U' \subseteq U \}.$	$\Omega(U) := \{ U' \in \mathbf{Op} \mid U' \subseteq U \}.$	$\Omega(U) := \{ U' \in \mathbf{Op} \mid U' \subseteq U \}.$
fong-spivak-seven-sketches_1 EQ0260	324	■ 0.996	$p ; h_3 = r ; g ; h_3 = r ; h_2 ; g' = q ; f' ; g'$	$p; h_3 = r; g; h_3 = r; h_2; g' = q; f'; g'$	$p \circ h_3 = r \circ g \circ h_3 = r \circ h_2 \circ g' = q \circ f' \circ g'$
fong-spivak-seven-sketches_1 EQ0088 (4.28)	140	■ 0.996	$\Phi(p, q) = I \otimes \Phi(p, q) \leq \mathcal{P}(p, p) \otimes \Phi(p, q) \leq \bigvee_{p_1 \in P} (\mathcal{P}(p, p_1) \otimes \Phi(p_1, q)) = (U_P; \Phi)(p, q).$	$\Phi(p, q) = I \otimes \Phi(p, q) \leq \mathcal{P}(p, p) \otimes \Phi(p, q) \leq \bigvee_{p_1 \in P} (\mathcal{P}(p, p_1) \otimes \Phi(p_1, q)) = (U_P; \Phi)(p, q).$	$\Phi(p, q) = I \otimes \Phi(p, q) \leq \mathcal{P}(p, p) \otimes \Phi(p, q) \leq \bigvee_{p_1 \in P} (\mathcal{P}(p, p_1) \otimes \Phi(p_1, q)) = (U_P \circ \Phi)(p, q).$
fong-spivak-seven-sketches_1 EQ0185	276	■ 0.996	$f^*(P) = \{ \{ x \in \mathbb{Z} \mid x \leq 0 \}, \{ x \in \mathbb{Z} \mid x \geq 1 \} \} \quad \text{and} \quad f^*(Q) = \{ \{ 0 \}, \{ x \in \mathbb{Z} \mid x \neq 0 \} \}.$	$f^*(P) = \{ \{ x \in \mathbb{Z} \mid x \leq 0 \}, \{ x \in \mathbb{Z} \mid x \geq 1 \} \} \quad \text{and} \quad f^*(Q) = \{ \{ 0 \}, \{ x \in \mathbb{Z} \mid x \neq 0 \} \}.$	$f^*(P) = \{ \{ x \in \mathbb{Z} \mid x \leq 0 \}, \{ x \in \mathbb{Z} \mid x \geq 1 \} \} \quad \text{and} \quad f^*(Q) = \{ \{ 0 \}, \{ x \in \mathbb{Z} \mid x \neq 0 \} \}.$
fong-spivak-seven-sketches_1 EQ0166 (7.57)	257	■ 0.996	$(U \Rightarrow V) := \bigcup_{\{ R \in \mathbf{Op} \mid R \cap U \subseteq V \}} R.$	$(U \Rightarrow V) := \bigcup_{\{ R \in \mathbf{Op} \mid R \cap U \subseteq V \}} R.$	$(U \Rightarrow V) := \bigcup_{\{ R \in \mathbf{Op} \mid R \cap U \subseteq V \}} R.$
fong-spivak-seven-sketches_1 EQ0013 (1.116)	44	■ 0.997	$f(p) := \bigwedge \{ q \in Q \mid p \leq g(q) \};$	$f(p) := \bigwedge \{ q \in Q \mid p \leq g(q) \};$	$f(p) := \bigwedge \{ q \in Q \mid p \leq g(q) \};$
fong-spivak-seven-sketches_1 EQ0154	243	■ 0.997	$\begin{array}{cc c c} P & Q & P \wedge Q & P = (P \wedge Q) \\ \hline \text{true} & \text{true} & ? & ? \\ \text{true} & \text{false} & ? & ? \\ \text{false} & \text{true} & ? & ? \\ \text{false} & \text{false} & ? & ? \end{array}$	$\begin{array}{cc c c} P & Q & P \wedge Q & P = (P \wedge Q) \\ \hline \text{true} & \text{true} & ? & ? \\ \text{true} & \text{false} & ? & ? \\ \text{false} & \text{true} & ? & ? \\ \text{false} & \text{false} & ? & ? \end{array}$	$\begin{array}{cc c c} P & Q & P \wedge Q & P = (P \wedge Q) \\ \hline \text{true} & \text{true} & ? & ? \\ \text{true} & \text{false} & ? & ? \\ \text{false} & \text{true} & ? & ? \\ \text{false} & \text{false} & ? & ? \end{array}$

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fong-spivak-seven-sketches_1 EQ0041	81	■ 0.998	$a+x \geq y \quad \text{\texttt{\text{iff}}} \quad a \geq y-x \quad \text{\texttt{\text{iff}}} \quad \max(0, a) \geq \max(0, y-x) \quad \text{\texttt{\text{iff}}} \quad a \geq (x \multimap y)$	$a+x \geq y \quad \text{\texttt{\text{iff}}} \quad a \geq y-x \quad \text{\texttt{\text{iff}}} \quad \max(0, a) \geq \max(0, y-x) \quad \text{\texttt{\text{iff}}} \quad a \geq (x \multimap y)$	$a+x \geq y \quad \text{\texttt{\text{iff}}} \quad a \geq y-x \quad \text{\texttt{\text{iff}}} \quad \max(0, a) \geq \max(0, y-x) \quad \text{\texttt{\text{iff}}} \quad a \geq (x \multimap y)$
fong-spivak-seven-sketches_1 EQ0082 (4.14)	135	■ 0.998	$\Phi(B, x)=11, \quad \Phi(A, z)=20, \quad \Phi(C, y)=17 \text{ .}$	$\Phi(B, x) = 11, \quad \Phi(A, z) = 20, \quad \Phi(C, y) = 17.$	$\Phi(B, x) = 11, \quad \Phi(A, z) = 20, \quad \Phi(C, y) = 17.$
fong-spivak-seven-sketches_1 EQ0124	190	■ 0.998	$v=\int \frac{1}{m} F(t) \, dt+u(t)+p \int a u(t)+b v(t) \, dt \text{ .}$	$v = \int \frac{1}{m} F(t) dt + u(t) + p \int a u(t) + b v(t) dt.$	$v = \int \frac{1}{m} F(t) dt + u(t) + p \int a u(t) + b v(t) dt.$
fong-spivak-seven-sketches_1 EQ0198	288	■ 0.998	$\begin{aligned} I_X * M(x, y) &= \bigvee_{x' \in X} I_X(x, x') \otimes M(x', y) \\ &= (I_X(x, x) \otimes M(x, y)) \vee \left(\bigvee_{x' \in X, x' \neq x} I_X(x, x') \otimes M(x', y) \right) \\ &= (I \otimes M(x, y)) \vee \left(\bigvee_{x' \in X, x' \neq x} 0 \otimes M(x', y) \right) \\ &= M(x, y) \vee 0 = M(x, y). \end{aligned}$	$\begin{aligned} I_X * M(x, y) &= \bigvee_{x' \in X} I_X(x, x') \otimes M(x', y) \\ &= (I_X(x, x) \otimes M(x, y)) \vee \left(\bigvee_{x' \in X, x' \neq x} I_X(x, x') \otimes M(x', y) \right) \\ &= (I \otimes M(x, y)) \vee \left(\bigvee_{x' \in X, x' \neq x} 0 \otimes M(x', y) \right) \\ &= M(x, y) \vee 0 = M(x, y). \end{aligned}$	$\begin{aligned} I_X * M(x, y) &= \bigvee_{x' \in X} I_X(x, x') \otimes M(x', y) \\ &= (I_X(x, x) \otimes M(x, y)) \vee \left(\bigvee_{x' \in X, x' \neq x} I_X(x, x') \otimes M(x', y) \right) \\ &= (I \otimes M(x, y)) \vee \left(\bigvee_{x' \in X, x' \neq x} 0 \otimes M(x', y) \right) \\ &= M(x, y) \vee 0 = M(x, y). \end{aligned}$
fong-spivak-seven-sketches_1 EQ0070	117	■ 0.999	$\Sigma_F(I)(\text{\texttt{Airline Seat}})=I(\text{\texttt{Economy}}) \sqcup I(\text{\texttt{First Class}}) \text{ .}$	$\Sigma_F(I)(\text{\texttt{Airline Seat}}) = I(\text{\texttt{Economy}}) \sqcup I(\text{\texttt{First Class}}) \text{ .}$	$\Sigma_F(I)(\text{\texttt{Airline Seat}}) = I(\text{\texttt{Economy}}) \sqcup I(\text{\texttt{First Class}}).$
fong-spivak-seven-sketches_1 EQ0084 (4.20)	138	■ 0.999	$(\Phi ; \Psi)(p, r):=\bigvee_{q \in Q} \Phi(p, q) \wedge \Psi(q, r) \text{ .}$	$(\Phi; \Psi)(p, r) := \bigvee_{q \in Q} \Phi(p, q) \wedge \Psi(q, r).$	$(\Phi \circ \Psi)(p, r) := \bigvee_{q \in Q} \Phi(p, q) \wedge \Psi(q, r).$
fong-spivak-seven-sketches_1 EQ0008 (1.96)	39	■ 0.999	$f(p) \leq q \quad \text{\texttt{\text{if and only if}}} \quad p \leq g(q) \text{ .}$	$f(p) \leq q \quad \text{\texttt{\text{if and only if}}} \quad p \leq g(q).$	$f(p) \leq q \quad \text{\texttt{\text{if and only if}}} \quad p \leq g(q).$
fong-spivak-seven-sketches_1 EQ0035	76	■ 0.999	$\begin{array}{l} I_W \leq f(I_V) \quad (f \text{ is monoidal monotone }) \\ \leq f(\mathcal{C}(c, c)) \quad (\mathcal{C} \text{ is } \mathcal{V}\text{-category}) \\ = \mathcal{C}_f(c, c) \quad (\text{definition of } \mathcal{C}_f) \end{array}$	$\begin{array}{l} I_W \leq f(I_V) \quad (f \text{ is monoidal monotone }) \\ \leq f(\mathcal{C}(c, c)) \quad (\mathcal{C} \text{ is } \mathcal{V}\text{-category}) \\ = \mathcal{C}_f(c, c) \quad (\text{definition of } \mathcal{C}_f) \end{array}$	$\begin{array}{l} I_W \leq f(I_V) \quad (f \text{ is monoidal monotone }) \\ \leq f(\mathcal{C}(c, c)) \quad (\mathcal{C} \text{ is } \mathcal{V}\text{-category}) \\ = \mathcal{C}_f(c, c) \quad (\text{definition of } \mathcal{C}_f) \end{array}$
fong-spivak-seven-sketches_1 EQ0042	82	■ 0.999	$a+c \rightarrow d \quad \text{\texttt{\text{if and only if}}} \quad a \rightarrow (c \rightarrow d) \text{ .}$	$a+c \rightarrow d \quad \text{\texttt{\text{if and only if}}} \quad a \rightarrow (c \rightarrow d).$	$a+c \rightarrow d \quad \text{\texttt{\text{if and only if}}} \quad a \rightarrow (c \multimap d).$
fong-spivak-seven-sketches_1 EQ0065 (3.66)	113	■ 0.999	$\text{\texttt{Gr}} \xrightarrow{F} \text{\texttt{DDS}} \xrightarrow{I} \text{\texttt{Set}} \text{ .}$	$\text{\texttt{Gr}} \xrightarrow{F} \text{\texttt{DDS}} \xrightarrow{I} \text{\texttt{Set}}.$	$\text{\texttt{Gr}} \xrightarrow{F} \text{\texttt{DDS}} \xrightarrow{I} \text{\texttt{Set}}.$
fong-spivak-seven-sketches_1 EQ0126 (6.15)	199	■ 0.999	$[f, g](x)= \begin{cases} f(x) & \text{\texttt{\text{if}}} \text{ } x=\iota_A(a) \text{ for some } a \in A; \\ g(x) & \text{\texttt{\text{if}}} \text{ } x=\iota_B(b) \text{ for some } b \in B \text{ .} \end{cases}$	$[f, g](x) = \begin{cases} f(x) & \text{\texttt{\text{if}}} \text{ } x = \iota_A(a) \text{ for some } a \in A; \\ g(x) & \text{\texttt{\text{if}}} \text{ } x = \iota_B(b) \text{ for some } b \in B. \end{cases}$	$[f, g](x) = \begin{cases} f(x) & \text{\texttt{\text{if}}} \text{ } x = \iota_A(a) \text{ for some } a \in A; \\ g(x) & \text{\texttt{\text{if}}} \text{ } x = \iota_B(b) \text{ for some } b \in B. \end{cases}$

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fong-spivak-seven-sketches EQ0139	220	■ 0.999	$\begin{aligned} \operatorname{Circ}(f): \operatorname{Circ}(V) &\longrightarrow \operatorname{Circ}(V') \\ (V, A, s, t, \ell) &\longmapsto (V', A, (s; f), (t; f), \ell). \end{aligned}$	$\operatorname{Circ}(f) : \operatorname{Circ}(V) \longrightarrow \operatorname{Circ}(V'); \\ (V, A, s, t, \ell) \longmapsto (V', A, (s; f), (t; f), \ell).$	$\operatorname{Circ}(f): \operatorname{Circ}(V) \longrightarrow \operatorname{Circ}(V'); \\ (V, A, s, t, \ell) \longmapsto (V', A, (s \circ f), (t \circ f), \ell).$
fong-spivak-seven-sketches EQ0032	68	■ 1.000	$d(x) := \begin{cases} \text{false} & \text{if } x > 0 \\ \text{true} & \text{if } x = 0 \end{cases} \quad u(x) := \begin{cases} \text{false} & \text{if } x = \infty \\ \text{true} & \text{if } x < \infty \end{cases}$	$d(x) := \begin{cases} \text{false} & \text{if } x > 0 \\ \text{true} & \text{if } x = 0 \end{cases} \quad u(x) := \begin{cases} \text{false} & \text{if } x = \infty \\ \text{true} & \text{if } x < \infty \end{cases}$	$d(x) := \begin{cases} \text{false} & \text{if } x > 0 \\ \text{true} & \text{if } x = 0 \end{cases} \quad u(x) := \begin{cases} \text{false} & \text{if } x = \infty \\ \text{true} & \text{if } x < \infty \end{cases}$
fong-spivak-seven-sketches EQ0176	272	■ 1.000	$\varnothing, \{1\}, \{2\}, \{3\}, \{1, 2\}, \{1, 3\}, \{2, 3\}, \{1, 2, 3\}.$	$\varnothing, \{1\}, \{2\}, \{3\}, \{1, 2\}, \{1, 3\}, \{2, 3\}, \{1, 2, 3\}.$	$\varnothing, \{1\}, \{2\}, \{3\}, \{1, 2\}, \{1, 3\}, \{2, 3\}, \{1, 2, 3\}.$
fong-spivak-seven-sketches EQ0252	313	■ 1.000	$\begin{aligned} \operatorname{B}(g^{\operatorname{op}}) &= \{(y, x) \in R^n \times R^m \mid y = S(g)(x)\} \\ \operatorname{B}(h) &= \{(x, z) \in R^m \times R^p \mid S(h)(x) = z\} \end{aligned}$	$\begin{aligned} \operatorname{B}(g^{\operatorname{op}}) &= \{(y, x) \in R^n \times R^m \mid y = S(g)(x)\} \\ \operatorname{B}(h) &= \{(x, z) \in R^m \times R^p \mid S(h)(x) = z\} \end{aligned}$	$\begin{aligned} \operatorname{B}(g^{\operatorname{op}}) &= \{(y, x) \in R^n \times R^m \mid y = S(g)(x)\} \\ \operatorname{B}(h) &= \{(x, z) \in R^m \times R^p \mid S(h)(x) = z\} \end{aligned}$
fong-spivak-seven-sketches EQ0001 (1.8)	18	■ 1.000	$\Phi(A) \vee \Phi(B) \leq \Phi(A \vee B).$	$\Phi(A) \vee \Phi(B) \leq \Phi(A \vee B).$	$\Phi(A) \vee \Phi(B) \leq \Phi(A \vee B).$
fong-spivak-seven-sketches EQ0004	24	■ 1.000	$f(11)=a, \quad \text{quad } f(12)=a, \quad \text{quad } f(13)=b, \quad \text{quad } f(21)=c, \\ \text{quad } f(22)=d, \quad \text{quad } f(23)=d$	$f(11) = a, \quad f(12) = a, \quad f(13) = b, \quad f(21) = c, \quad f(22) = d, \quad f(23) = d$	$f(11) = a, \quad f(12) = a, \quad f(13) = b, \quad f(21) = c, \quad f(22) = d, \quad f(23) = d$
fong-spivak-seven-sketches EQ0006	35	■ 1.000	$\left\{ \frac{1}{n+1} \mid n \in \mathbb{N} \right\} = \left\{ 1, \frac{1}{2}, \frac{1}{3}, \frac{1}{4}, \dots \right\} \subseteq \mathbb{R}$	$\left\{ \frac{1}{n+1} \mid n \in \mathbb{N} \right\} = \left\{ 1, \frac{1}{2}, \frac{1}{3}, \frac{1}{4}, \dots \right\} \subseteq \mathbb{R}$	$\left\{ \frac{1}{n+1} \mid n \in \mathbb{N} \right\} = \left\{ 1, \frac{1}{2}, \frac{1}{3}, \frac{1}{4}, \dots \right\} \subseteq \mathbb{R}$
fong-spivak-seven-sketches EQ0009	39	■ 1.000	$\lceil x / 3 \rceil \leq y \text{ if and only if } x \leq 3y.$	$\lceil x/3 \rceil \leq y \text{ if and only if } x \leq 3y.$	$\lceil x/3 \rceil \leq y \text{ if and only if } x \leq 3y.$
fong-spivak-seven-sketches EQ0011	43	■ 1.000	$g(\bigwedge A) \cong \bigwedge g(A).$	$g(\bigwedge A) \cong \bigwedge g(A).$	$g\left(\bigwedge A\right) \cong \bigwedge g(A).$
fong-spivak-seven-sketches EQ0012	43	■ 1.000	$f(\bigvee A) \cong \bigvee f(A)$	$f(\bigvee A) \cong \bigvee f(A)$	$f\left(\bigvee A\right) \cong \bigvee f(A).$
fong-spivak-seven-sketches EQ0014	44	■ 1.000	$p_{\{0\}} \leq \bigwedge \{g(q) \in P \mid p_{\{0\}} \leq g(q)\} \cong g\left(\bigwedge \{q \in Q \mid p_{\{0\}} \leq g(q)\}\right) = g(f(p_0)),$	$p_0 \leq \bigwedge \{g(q) \in P \mid p_0 \leq g(q)\} \cong g\left(\bigwedge \{q \in Q \mid p_0 \leq g(q)\}\right) = g(f(p_0)),$	$p_0 \leq \bigwedge \{g(q) \in P \mid p_0 \leq g(q)\} \cong g\left(\bigwedge \{q \in Q \mid p_0 \leq g(q)\}\right) = g(f(p_0)),$
fong-spivak-seven-sketches EQ0015	44	■ 1.000	$f\left(g\left(\bigwedge_{q \in Q} q\right)\right) = \bigwedge_{q \in Q} f(g(q)) \leq \bigwedge_{q \in Q} q = q_0,$	$f(g(q_0)) = \bigwedge \{q \in Q \mid g(q_0) \leq g(q)\} \leq \bigwedge \{q_0\} = q_0,$	$f(g(q_0)) = \bigwedge \{q \in Q \mid g(q_0) \leq g(q)\} \leq \bigwedge \{q_0\} = q_0,$
fong-spivak-seven-sketches EQ0016	45	■ 1.000	$f_!(A') := \{b \in B \mid \text{there exists } a \in A' \text{ such that } f(a) = b\}$	$f_!(A') := \{b \in B \mid \text{there exists } a \in A' \text{ such that } f(a) = b\}$	$f_!(A') := \{b \in B \mid \text{there exists } a \in A' \text{ such that } f(a) = b\}$
fong-spivak-seven-sketches EQ0017	45	■ 1.000	$f_*(A') := \{b \in B \mid \text{for all } a \text{ such that } f(a) = b, \text{ we have } a \in A'\}$	$f_*(A') := \{b \in B \mid \text{for all } a \text{ such that } f(a) = b, \text{ we have } a \in A'\}$	$f_*(A') := \{b \in B \mid \text{for all } a \text{ such that } f(a) = b, \text{ we have } a \in A'\}$
fong-spivak-seven-sketches EQ0019 (2.7)	54	■ 1.000	$m * e = m, \quad e * m = m, \quad (m * n) * p = m * (n * p)$	$m * e = m, \quad e * m = m, \quad (m * n) * p = m * (n * p)$	$m * e = m, \quad e * m = m, \quad (m * n) * p = m * (n * p)$

Identifier	Page	Conf.	LaTeX source	Rendered	Image
fong-spivak-seven-sketches_1EQ0052	87	■ 1.000	$M_{\{Y\}^{\wedge 3}}=\begin{array}{c ccc}\nearrow & x & y & z \\ \hline x & 0 & 4 & 3 \\ y & 3 & 0 & 6 \\ z & 7 & 4 & 0 \end{array}$	$M_Y^3 = \begin{array}{c ccc}\nearrow & x & y & z \\ \hline x & 0 & 4 & 3 \\ y & 3 & 0 & 6 \\ z & 7 & 4 & 0 \end{array}$	$M_Y^3 = \begin{array}{c ccc}\nearrow & x & y & z \\ \hline x & 0 & 4 & 3 \\ y & 3 & 0 & 6 \\ z & 7 & 4 & 0 \end{array}$
fong-spivak-seven-sketches_1EQ0055	96	■ 1.000	$\{A, B, C, D, f, g, h, i, f ; h\}$	$\{A, B, C, D, f, g, h, i, f; h\}$	$\{A, B, C, D, f, g, h, i, f \circ h\}$
fong-spivak-seven-sketches_1EQ0058	104	■ 1.000	$f^{\wedge \{ \prime \}} ; h^{\wedge \{ \prime \}} = F(f) ; F(h) = F(f ; h) = F(g ; i) = F(g) ; F(i) = g^{\wedge \{ \prime \}} ; i^{\wedge \{ \prime \}}$	$f'; h' = F(f); F(h) = F(f; h) = F(g; i) = F(g); F(i) = g'; i'$	$f' \circ h' = F(f) \circ F(h) = F(f \circ h) = F(g \circ i) = F(g) \circ F(i) = g' \circ i'$
fong-spivak-seven-sketches_1EQ0060 (3.50)	107	■ 1.000	$F(f) ; \alpha_d = \alpha_c ; G(f) .$	$F(f); \alpha_d = \alpha_c; G(f).$	$F(f) \circ \alpha_d = \alpha_c \circ G(f).$
fong-spivak-seven-sketches_1EQ0062	110	■ 1.000	$\alpha_{\text{\text{Vertex}}}: G(\text{\text{Vertex}}) \rightarrow H(\text{\text{Vertex}}) \quad \text{and} \quad \alpha_{\text{\text{Arrow}}}: G(\text{\text{Arrow}}) \rightarrow H(\text{\text{Arrow}}),$	$\alpha_{\text{Vertex}}: G(\text{Vertex}) \rightarrow H(\text{Vertex}) \quad \text{and} \quad \alpha_{\text{Arrow}}: G(\text{Arrow}) \rightarrow H(\text{Arrow}),$	$\alpha_{\text{Vertex}}: G(\text{Vertex}) \rightarrow H(\text{Vertex}) \quad \text{and} \quad \alpha_{\text{Arrow}}: G(\text{Arrow}) \rightarrow H(\text{Arrow}),$
fong-spivak-seven-sketches_1EQ0067	114	■ 1.000	$\alpha_{c,d}: \mathcal{C}(c, R(d)) \xrightarrow{\cong} \mathcal{D}(L(c), d)$	$\alpha_{c,d}: \mathcal{C}(c, R(d)) \xrightarrow{\cong} \mathcal{D}(L(c), d)$	$\alpha_{c,d}: \mathcal{C}(c, R(d)) \xrightarrow{\cong} \mathcal{D}(L(c), d)$
fong-spivak-seven-sketches_1EQ0076	121	■ 1.000	$X \times Y := \{(x, y) \mid x \in X, y \in Y\},$	$X \times Y := \{(x, y) \mid x \in X, y \in Y\},$	$X \times Y := \{(x, y) \mid x \in X, y \in Y\},$
fong-spivak-seven-sketches_1EQ0090	141	■ 1.000	$(\Phi ; \Psi) ; \Upsilon = \Phi ; (\Psi ; \Upsilon) .$	$(\Phi; \Psi); \Upsilon = \Phi; (\Psi; \Upsilon).$	$(\Phi \circ \Psi) \circ \Upsilon = \Phi \circ (\Psi \circ \Upsilon).$
fong-spivak-seven-sketches_1EQ0095	148	■ 1.000	$\alpha_{s,t,u}: \{(s,(t,u)) \mid s \in S, t \in T, u \in U\} \xrightarrow{\cong} \{((s,t),u) \mid s \in S, t \in T, u \in U\} .$	$\alpha_{s,t,u}: \{(s,(t,u)) \mid s \in S, t \in T, u \in U\} \xrightarrow{\cong} \{((s,t),u) \mid s \in S, t \in T, u \in U\} .$	$\alpha_{s,t,u}: \{(s,(t,u)) \mid s \in S, t \in T, u \in U\} \xrightarrow{\cong} \{((s,t),u) \mid s \in S, t \in T, u \in U\} .$
fong-spivak-seven-sketches_1EQ0099	156	■ 1.000	$(\Phi \times \Psi) (\left(\left(x_1\right), \left(y_1\right)\right), \left(x_2\right), \left(y_2\right)) := \Phi \left(x_1\right), \left(x_2\right) \times \Psi \left(y_1\right), \left(y_2\right) .$	$(\Phi \times \Psi) ((x_1, y_1), (x_2, y_2)) := \Phi(x_1, x_2) \otimes \Psi(y_1, y_2) .$	$(\Phi \times \Psi)((x_1, y_1), (x_2, y_2)) := \Phi(x_1, x_2) \otimes \Psi(y_1, y_2).$
fong-spivak-seven-sketches_1EQ0101 (5.4)	162	■ 1.000	$i \longmapsto \begin{cases} f(i) & \text{if } 1 \leq i \leq m \\ m^{\prime}+g(i) & \text{if } m+1 \leq i \leq m+n \end{cases}$	$i \mapsto \begin{cases} f(i) & \text{if } 1 \leq i \leq m; \\ m' + g(i) & \text{if } m + 1 \leq i \leq m + n. \end{cases}$	$i \mapsto \begin{cases} f(i) & \text{if } 1 \leq i \leq m; \\ m' + g(i) & \text{if } m + 1 \leq i \leq m + n. \end{cases}$
fong-spivak-seven-sketches_1EQ0106 (5.26)	168	■ 1.000	$G:=\left\{\rho_{m,n} \mid m,n \in \mathbb{N}\right\}, \quad \quad \quad s\left(\rho_{m,n}\right):=m, \quad \quad t\left(\rho_{m,n}\right):=n,$	$G := \{\rho_{m,n} \mid m,n \in \mathbb{N}\}, \quad s(\rho_{m,n}) := m, \quad t(\rho_{m,n}) := n,$	$G := \{\rho_{m,n} \mid m,n \in \mathbb{N}\}, \quad s(\rho_{m,n}) := m, \quad t(\rho_{m,n}) := n,$
fong-spivak-seven-sketches_1EQ0108	174	■ 1.000	$\begin{array}{r} D \ x+3 \ D^2 \ y-2 \ z=0 \ \ \ D^2 \ y+5 \ D \ z=0 \\ \end{array}$	$\begin{array}{l} Dx+3D^2y-2z=0 \\ D^2y+5Dz=0 \end{array}$	$\begin{array}{l} Dx+3D^2y-2z=0 \\ D^2y+5Dz=0 \end{array}$
fong-spivak-seven-sketches_1EQ0110 (5.48)	176	■ 1.000	$M ; N(a, c) := \sum_{b \in \underline{n}} M(a, b) \times N(b, c),$	$M; N(a, c) := \sum_{b \in \underline{n}} M(a, b) \times N(b, c),$	$M \circ N(a, c) := \sum_{b \in \underline{n}} M(a, b) \times N(b, c),$
fong-spivak-seven-sketches_1EQ0112	176	■ 1.000	$A=\left(\begin{array}{lll} 3 & 3 & 1 \\ 2 & 0 & 4 \end{array}\right) \quad B=\left(\begin{array}{llll} 2 & 5 & 6 & 1 \end{array}\right) .$	$A = \left(\begin{array}{ccc} 3 & 3 & 1 \\ 2 & 0 & 4 \end{array} \right) \quad B = \left(\begin{array}{cccc} 2 & 5 & 6 & 1 \end{array} \right) .$	$A = \left(\begin{array}{ccc} 3 & 3 & 1 \\ 2 & 0 & 4 \end{array} \right) \quad B = \left(\begin{array}{cccc} 2 & 5 & 6 & 1 \end{array} \right) .$
fong-spivak-seven-sketches_1EQ0114	179	■ 1.000	$\sum_{j \in \underline{n}} \alpha(i, j) * \beta(j, k) .$	$\sum_{j \in \underline{n}} \alpha(i, j) * \beta(j, k).$	$\sum_{j \in \underline{n}} \alpha(i, j) * \beta(j, k).$

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fong-spivak-seven-sketches- EQ0118 (5.76)	186	■ 1.000	$\mathrm{B}\backslash\left(g^{\mathrm{op}}\backslash\right):=\backslash\left\{\left(S(g)(x), x\right) \mid x \in R^{\mathrm{m}}\backslash\right\} \subseteq R^{\mathrm{n}} \times R^{\mathrm{m}}$.	$B\left(g^{\mathrm{op}}\right):=\left\{\left(S(g)(x), x\right) \mid x \in R^m\right\} \subseteq R^n \times R^m$.	$B\left(g^{\mathrm{op}}\right):=\left\{\left(S(g)(x), x\right) \mid x \in R^m\right\} \subseteq R^n \times R^m$.
fong-spivak-seven-sketches- EQ0122	188	■ 1.000	$\mathrm{B}\backslash\left(\backslash\left(g^{\mathrm{op}}\backslash\right) ; h\backslash\right)=\backslash\left\{\left(S(g)(x), S(h)(x)\right) \mid x \in R^{\mathrm{m}}\backslash\right\}$.	$B\left(\left(g^{\mathrm{op}}\right) ; h\right)=\left\{\left(S(g)(x), S(h)(x)\right) \mid x \in R^m\right\}$.	$B\left(\left(g^{\mathrm{op}}\right) ; h\right)=\left\{\left(S(g)(x), S(h)(x)\right) \mid x \in R^m\right\}$.
fong-spivak-seven-sketches- EQ0127	203	■ 1.000	$y(0)=y(g(0))=x(f(0))=x(0)$.	$y(0)=y(g(0))=x(f(0))=x(0)$.	$y(0)=y(g(0))=x(f(0))=x(0)$.
fong-spivak-seven-sketches- EQ0130	206	■ 1.000	$\{(v, d) \mid d \in D(v)\}$	$\{(v, d) \mid d \in D(v)\}$	$\{(v, d) \mid d \in D(v)\}$
fong-spivak-seven-sketches- EQ0135 (6.76)	219	■ 1.000	$F\backslash\left(\backslash\left[\iota_{\mathrm{N}}, \iota_{\mathrm{P}}\backslash\right]\backslash\right)\backslash\left(\backslash\mathrm{varphi}_{\mathrm{N}, \mathrm{P}}(s, t)\backslash\right)$	$F\left([\iota_N, \iota_P]\right)\left(\varphi_{N, P}(s, t)\right)$	$F\left([\iota_N, \iota_P]\right)\left(\varphi_{N, P}(s, t)\right)$
fong-spivak-seven-sketches- EQ0138	220	■ 1.000	$\mathrm{operatorname{Circ}}(V):=\{(V, A, s, t, \ell) \mid \text { where } s, t: A \rightarrow V, \ell: E \rightarrow C\}$.	$\mathrm{Circ}(V):=\{(V, A, s, t, \ell) \mid \text { where } s, t: A \rightarrow V, \ell: E \rightarrow C\}$.	$\mathrm{Circ}(V):=\{(V, A, s, t, \ell) \mid \text { where } s, t: A \rightarrow V, \ell: E \rightarrow C\}$.
fong-spivak-seven-sketches- EQ0141	225	■ 1.000	$\mathrm{circ}_{\mathrm{i}}: \mathcal{O}\backslash\left(s_{\mathrm{1}}, \ldots, s_{\mathrm{m}} ; t_{\mathrm{i}}\backslash\right) \times \mathcal{O}\backslash\left(t_{\mathrm{1}}, \ldots, t_{\mathrm{n}}\backslash\right) \rightarrow \mathcal{O}\backslash\left(s_{\mathrm{1}}, \ldots, s_{\mathrm{m}}, t_{\mathrm{i}+1}, \ldots, t_{\mathrm{n}} ; t\backslash\right) ;$	$\circ_i: \mathcal{O}\left(s_1, \ldots, s_m ; t_i\right) \times \mathcal{O}\left(t_1, \ldots, t_n ; t\right) \rightarrow \mathcal{O}\left(t_1, \ldots, t_{i-1}, s_1, \ldots, s_m, t_{i+1}, \ldots, t_n ; t\right) ;$	$\circ_i: \mathcal{O}\left(s_1, \ldots, s_m ; t_i\right) \times \mathcal{O}\left(t_1, \ldots, t_n ; t\right) \rightarrow \mathcal{O}\left(t_1, \ldots, t_{i-1}, s_1, \ldots, s_m, t_{i+1}, \ldots, t_n ; t\right) ;$
fong-spivak-seven-sketches- EQ0145	229	■ 1.000	$F: \mathcal{O}\backslash\left(t_{\mathrm{1}}, \ldots, t_{\mathrm{n}} ; t\backslash\right) \rightarrow \mathcal{P}\backslash\left(f\backslash\left(t_{\mathrm{1}}\backslash\right), \ldots, f\backslash\left(t_{\mathrm{n}}\backslash\right) ; f(t)\backslash\right)$	$F: \mathcal{O}\left(t_1, \ldots, t_n ; t\right) \rightarrow \mathcal{P}\left(f\left(t_1\right), \ldots, f\left(t_n\right) ; f(t)\right)$	$F: \mathcal{O}\left(t_1, \ldots, t_n ; t\right) \rightarrow \mathcal{P}\left(f\left(t_1\right), \ldots, f\left(t_n\right) ; f(t)\right)$
fong-spivak-seven-sketches- EQ0149 (7.10)	239	■ 1.000	$\mathcal{C}(A \times C, D) \cong \mathcal{C}\backslash\left(A, D^{\mathrm{C}}\backslash\right)$	$\mathcal{C}(A \times C, D) \cong \mathcal{C}\left(A, D^C\right)$	$\mathcal{C}(A \times C, D) \cong \mathcal{C}(A, D^C)$
fong-spivak-seven-sketches- EQ0151	241	■ 1.000	$\ulcorner m\urcorner(y):=\begin{cases} \text { true } & \& \text { if } m(x)=y \text { for some } x \in X \\ \text { false } & \& \text { otherwise } \end{cases}$	$\ulcorner m\urcorner(y):=\begin{cases} \text { true } & \text { if } m(x)=y \text { for some } x \in X \\ \text { false } & \text { otherwise } \end{cases}$	$\ulcorner m\urcorner(y):=\begin{cases} \text { true } & \text { if } m(x)=y \text { for some } x \in X \\ \text { false } & \text { otherwise } \end{cases}$
fong-spivak-seven-sketches- EQ0152 (7.15)	241	■ 1.000	$X=\{y \in Y \mid p(y)=\text { true }\}$.	$X=\{y \in Y \mid p(y)=\text { true }\}$.	$X=\{y \in Y \mid p(y)=\text { true }\}$.
fong-spivak-seven-sketches- EQ0155	246	■ 1.000	$B(p ; \epsilon):=\left\{p^{\prime} \in \mathbb{R}^2 \mid d\left(p, p^{\prime}\right)<\epsilon\right\}^7$	$B(p ; \epsilon):=\left\{p' \in \mathbb{R}^2 \mid d(p, p') < \epsilon\right\}^7$	$B(p ; \epsilon):=\left\{p' \in \mathbb{R}^2 \mid d(p, p') < \epsilon\right\}^7$
fong-spivak-seven-sketches- EQ0161 (7.51)	255	■ 1.000	$\begin{aligned} & \Omega(U) \& \rightarrow \Omega(V) \backslash \backslash U^{\prime} \\ & U^{\prime} \mapsto U^{\prime} \cap V . \end{aligned}$	$\Omega(U) \rightarrow \Omega(V)$ $U' \mapsto U' \cap V$.	$\Omega(U) \rightarrow \Omega(V)$ $U' \mapsto U' \cap V$.
fong-spivak-seven-sketches- EQ0162	256	■ 1.000	$V \cap U_{\mathrm{j}}=\left(\bigcup_{i \in I} V_{\mathrm{i}}\right) \cap U_{\mathrm{j}}=\bigcup_{i \in I}\left(V_{\mathrm{i}} \cap U_{\mathrm{j}}\right)=\bigcup_{i \in I}\left(V_{\mathrm{j}} \cap U_{\mathrm{i}}\right)=\left(\bigcup_{i \in I} U_{\mathrm{i}}\right) \cap V_{\mathrm{j}}=V_{\mathrm{j}}$	$V \cap U_j = \left(\bigcup_{i \in I} V_i\right) \cap U_j = \bigcup_{i \in I} (V_i \cap U_j) = \bigcup_{i \in I} (V_j \cap U_i) = \left(\bigcup_{i \in I} U_i\right) \cap V_j = V_j$	$V \cap U_j = \left(\bigcup_{i \in I} V_i\right) \cap U_j = \bigcup_{i \in I} (V_i \cap U_j) = \bigcup_{i \in I} (V_j \cap U_i) = \left(\bigcup_{i \in I} U_i\right) \cap V_j = V_j$
fong-spivak-seven-sketches- EQ0167 (7.63)	259	■ 1.000	$\left \Omega^S\right , \leq^S$	$(\left \Omega^S\right , \leq^S)$	$(\left \Omega^S\right , \leq^S)$
fong-spivak-seven-sketches- EQ0170	261	■ 1.000	$p(n, z)=\begin{cases} \text { true } & \text { if } n \leq z \\ \text { false } & \text { if } n> z . \end{cases}$	$p(n, z)=\begin{cases} \text { true } & \text { if } n \leq z \\ \text { false } & \text { if } n> z . \end{cases}$	$p(n, z)=\begin{cases} \text { true } & \text { if } n \leq z \\ \text { false } & \text { if } n> z . \end{cases}$
fong-spivak-seven-sketches- EQ0174	265	■ 1.000	$S\backslash\left(o_{[a, b]}\backslash\right) \cong \lim _{\epsilon>0} S \backslash\left(o_{[a-\epsilon, b+\epsilon]}\backslash\right)$.	$S\left(o_{[a, b]}\right) \cong \lim _{\epsilon>0} S\left(o_{[a-\epsilon, b+\epsilon]}\right)$.	$S\left(o_{[a, b]}\right) \cong \lim _{\epsilon>0} S\left(o_{[a-\epsilon, b+\epsilon]}\right)$.

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fong-spivak-seven-sketchesEQ0177	272	■1.000	$(h, 1), \quad (h, 2), \quad (h, 3), \quad (1, 1), \quad (1, 2), \quad (1, 3), \quad (1, 3) \quad .$	$(h, 1), \quad (h, 2), \quad (h, 3), \quad (1, 1), \quad (1, 2), \quad (1, 3).$	$(h, 1), \quad (h, 2), \quad (h, 3), \quad (1, 1), \quad (1, 2), \quad (1, 3).$
fong-spivak-seven-sketchesEQ0178	272	■1.000	$(h, 1), \quad (1, 1), \quad (1, 2), \quad (2, 2), \quad (3, 2) \quad .$	$(h, 1), \quad (1, 1), \quad (1, 2), \quad (2, 2), \quad (3, 2).$	$(h, 1), \quad (1, 1), \quad (1, 2), \quad (2, 2), \quad (3, 2).$
fong-spivak-seven-sketchesEQ0180	273	■1.000	$\begin{array}{lllll} (11, 11) & (11, 12) & (12, 11) & (12, 12) & (13, 13) \\ (12, 12) & (13, 13) & (21, 21) & (22, 22) & (12, 23) & (23, 22) & (23, 23) \end{array}$	$\begin{array}{lllll} (11, 11) & (11, 12) & (12, 11) & (12, 12) & (13, 13) \\ (21, 21) & (22, 22) & (12, 23) & (23, 22) & (23, 23) \end{array}$	$\begin{array}{lllll} (11, 11) & (11, 12) & (12, 11) & (12, 12) & (13, 13) \\ (21, 21) & (22, 22) & (12, 23) & (23, 22) & (23, 23) \end{array}$
fong-spivak-seven-sketchesEQ0182	274	■1.000	$1 \leq 1, \quad 1 \leq 2, \quad 1 \leq 3, \quad 2 \leq 2, \quad 2 \leq 3, \quad 3 \leq 3, \quad 4 \leq 4$	$1 \leq 1, \quad 1 \leq 2, \quad 1 \leq 3, \quad 2 \leq 2, \quad 2 \leq 3, \quad 3 \leq 3, \quad 4 \leq 4$	$1 \leq 1, \quad 1 \leq 2, \quad 1 \leq 3, \quad 2 \leq 2, \quad 2 \leq 3, \quad 3 \leq 3, \quad 4 \leq 4$
fong-spivak-seven-sketchesEQ0183	274	■1.000	$(\bullet)(\circ)(*), \quad (\bullet\circ)(*), \quad (\bullet*)(\circ), \quad (\bullet)(\circ*), \quad (\bullet\circ*)$	$(\bullet)(\circ)(*), \quad (\bullet\circ)(*), \quad (\bullet*)(\circ), \quad (\bullet)(\circ*), \quad (\bullet\circ*)$	$(\bullet)(\circ)(*), \quad (\bullet\circ)(*), \quad (\bullet*)(\circ), \quad (\bullet)(\circ*), \quad (\bullet\circ*)$
fong-spivak-seven-sketchesEQ0190	280	■1.000	$g(q) \leq g^{\prime}(f(g(q))) \leq g^{\prime}(q) \quad .$	$g(q) \leq g^{\prime}(f(g(q))) \leq g^{\prime}(q).$	$g(q) \leq g^{\prime}(f(g(q))) \leq g^{\prime}(q).$
fong-spivak-seven-sketchesEQ0192	283	■1.000	$\begin{gathered} g(\text{false})+g(\text{false})=\infty+\infty\geq\infty=g(\text{false}\wedge\text{false}) \\ g(\text{false})+g(\text{true})=\infty+0\geq\infty=g(\text{false}\wedge\text{true}) \\ g(\text{true})+g(\text{false})=0+\infty\geq\infty=g(\text{true}\wedge\text{false}) \\ g(\text{true})+g(\text{true})=0+0\geq0=g(\text{true}\wedge\text{true}). \end{gathered}$	$\begin{gathered} g(\text{false})+g(\text{false})=\infty+\infty\geq\infty=g(\text{false}\wedge\text{false}) \\ g(\text{false})+g(\text{true})=\infty+0\geq\infty=g(\text{false}\wedge\text{true}) \\ g(\text{true})+g(\text{false})=0+\infty\geq\infty=g(\text{true}\wedge\text{false}) \\ g(\text{true})+g(\text{true})=0+0\geq0=g(\text{true}\wedge\text{true}) \end{gathered}$	$\begin{gathered} g(\text{false})+g(\text{false})=\infty+\infty\geq\infty=g(\text{false}\wedge\text{false}) \\ g(\text{false})+g(\text{true})=\infty+0\geq\infty=g(\text{false}\wedge\text{true}) \\ g(\text{true})+g(\text{false})=0+\infty\geq\infty=g(\text{true}\wedge\text{false}) \\ g(\text{true})+g(\text{true})=0+0\geq0=g(\text{true}\wedge\text{true}). \end{gathered}$
fong-spivak-seven-sketchesEQ0196	287	■1.000	$\left(\begin{array}{ll} 1 & 0 \\ 0 & 1 \end{array}\right), \quad \left(\begin{array}{ll} \text{true} & \text{false} \\ \text{false} & \text{true} \end{array}\right), \quad \text{and} \quad \left(\begin{array}{ll} 0 & \infty \\ \infty & 0 \end{array}\right).$	$\left(\begin{array}{ll} 1 & 0 \\ 0 & 1 \end{array}\right), \quad \left(\begin{array}{ll} \text{true} & \text{false} \\ \text{false} & \text{true} \end{array}\right), \quad \text{and} \quad \left(\begin{array}{ll} 0 & \infty \\ \infty & 0 \end{array}\right).$	$\left(\begin{array}{ll} 1 & 0 \\ 0 & 1 \end{array}\right), \quad \left(\begin{array}{ll} \text{true} & \text{false} \\ \text{false} & \text{true} \end{array}\right), \quad \text{and} \quad \left(\begin{array}{ll} 0 & \infty \\ \infty & 0 \end{array}\right).$
fong-spivak-seven-sketchesEQ0197	287	■1.000	$0 \otimes v \cong v \otimes 0 \cong \left(v \otimes \bigvee_{a \in \emptyset} a\right) = \bigvee_{a \in \emptyset} (v \otimes a) = 0.$	$0 \otimes v \cong v \otimes 0 \cong \left(v \otimes \bigvee_{a \in \emptyset} a\right) = \bigvee_{a \in \emptyset} (v \otimes a) = 0.$	$0 \otimes v \cong v \otimes 0 \cong \left(v \otimes \bigvee_{a \in \emptyset} a\right) = \bigvee_{a \in \emptyset} (v \otimes a) = 0.$
fong-spivak-seven-sketchesEQ0200	288	■1.000	$M_X=\left(\begin{array}{cccc} 0 & \infty & 3 & \infty \\ 2 & 0 & \infty & 5 \\ \infty & 3 & 0 & \infty \\ \infty & \infty & 6 & 0 \end{array}\right) \quad M_X^2=\left(\begin{array}{cccc} 0 & 6 & 3 & \infty \\ 2 & 0 & 5 & 5 \\ 5 & 3 & 0 & 8 \\ \infty & 9 & 6 & 0 \end{array}\right) \quad M_X^3=M_X^4=\left(\begin{array}{cccc} 0 & 6 & 3 & 11 \\ 2 & 0 & 5 & 5 \\ 5 & 3 & 0 & 8 \\ 11 & 9 & 6 & 0 \end{array}\right)$	$M_X=\left(\begin{array}{cccc} 0 & \infty & 3 & \infty \\ 2 & 0 & \infty & 5 \\ \infty & 3 & 0 & \infty \\ \infty & \infty & 6 & 0 \end{array}\right) \quad M_X^2=\left(\begin{array}{cccc} 0 & 6 & 3 & \infty \\ 2 & 0 & 5 & 5 \\ 5 & 3 & 0 & 8 \\ \infty & 9 & 6 & 0 \end{array}\right) \quad M_X^3=M_X^4=\left(\begin{array}{cccc} 0 & 6 & 3 & 11 \\ 2 & 0 & 5 & 5 \\ 5 & 3 & 0 & 8 \\ 11 & 9 & 6 & 0 \end{array}\right)$	$M_X=\left(\begin{array}{cccc} 0 & \infty & 3 & \infty \\ 2 & 0 & \infty & 5 \\ \infty & 3 & 0 & \infty \\ \infty & \infty & 6 & 0 \end{array}\right) \quad M_X^2=\left(\begin{array}{cccc} 0 & 6 & 3 & \infty \\ 2 & 0 & 5 & 5 \\ 5 & 3 & 0 & 8 \\ \infty & 9 & 6 & 0 \end{array}\right) \quad M_X^3=M_X^4=\left(\begin{array}{cccc} 0 & 6 & 3 & 11 \\ 2 & 0 & 5 & 5 \\ 5 & 3 & 0 & 8 \\ 11 & 9 & 6 & 0 \end{array}\right)$
fong-spivak-seven-sketchesEQ0201	289	■1.000	$t(p):=\left(\begin{array}{l} v \text{ if } n=0 \\ t(a_n) \text{ if } n \geq 1 \end{array}\right)$	$t(p):=\left\{\begin{array}{l} v \text{ if } n=0 \\ t(a_n) \text{ if } n \geq 1 \end{array}\right.$	$t(p):=\left\{\begin{array}{l} v \text{ if } n=0 \\ t(a_n) \text{ if } n \geq 1 \end{array}\right.$
fong-spivak-seven-sketchesEQ0202	289	■1.000	$(p \circ q):=\left(v, a_1, \ldots, a_m, b_1, \ldots, b_n\right) \quad .$	$(p; q):=\left(v, a_1, \ldots, a_m, b_1, \ldots, b_n\right) \quad .$	$(p \circ q):=\left(v, a_1, \ldots, a_m, b_1, \ldots, b_n\right) \quad .$

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fong-spivak-seven-sketchesEQ0203	289	■ 1.000	$\left(v, a_1, \ldots, a_m, b_1, \ldots, b_n, c_1, \ldots, c_o, c_{\{1\}}, \ldots, c_{\{o\}}\right)$.	$(v, a_1, \ldots, a_m, b_1, \ldots, b_n, c_1, \ldots, c_o)$.	$(v, a_1, \ldots, a_m, b_1, \ldots, b_n, c_1, \ldots, c_o)$.
fong-spivak-seven-sketchesEQ0207	291	■ 1.000	$\begin{array}{lll} (1,1) & \& (1,2) & \& (1,3) \\ (2,1) & \& (2,2) & \& (2,3) \\ (3,1) & \& (3,2) & \& (3,3) \end{array}$	$\begin{array}{lll} (1,1) & (1,2) & (1,3) \\ (2,1) & (2,2) & (2,3) \\ (3,1) & (3,2) & (3,3) \end{array}$	$\begin{array}{lll} (1,1) & (1,2) & (1,3) \\ (2,1) & (2,2) & (2,3) \\ (3,1) & (3,2) & (3,3) \end{array}$
fong-spivak-seven-sketchesEQ0208	291	■ 1.000	$f^{-1}(1)=b, \quad f^{-1}(2)=a, \quad f^{-1}(3)=c$.	$f^{-1}(1)=b, \quad f^{-1}(2)=a, \quad f^{-1}(3)=c$.	$f^{-1}(1)=b, \quad f^{-1}(2)=a, \quad f^{-1}(3)=c$.
fong-spivak-seven-sketchesEQ0209	291	■ 1.000	$\begin{gathered} A^{\{\prime\}} \mapsto A, \quad f^{\{\prime\}} \mapsto f, \quad g^{\{\prime\}} \mapsto g, \quad h^{\{\prime\}} \mapsto h, \quad i^{\{\prime\}} \mapsto i, \\ B^{\{\prime\}} \mapsto B, \quad h^{\{\prime\}} \mapsto h, \quad C^{\{\prime\}} \mapsto C, \quad i^{\{\prime\}} \mapsto i, \quad D^{\{\prime\}} \mapsto D. \end{gathered}$	$A' \mapsto A, \quad f' \mapsto f, \quad g' \mapsto g, \quad f'; h' \mapsto f; h, \quad g'; i' \mapsto f; h, \\ B' \mapsto B, \quad h' \mapsto h, \quad C' \mapsto C, \quad i' \mapsto i, \quad D' \mapsto D.$	$A' \mapsto A, \quad f' \mapsto f, \quad g' \mapsto g, \quad f' \circ h' \mapsto f \circ h, \quad g' \circ i' \mapsto f \circ h, \\ B' \mapsto B, \quad h' \mapsto h, \quad C' \mapsto C, \quad i' \mapsto i, \quad D' \mapsto D.$
fong-spivak-seven-sketchesEQ0211	292	■ 1.000	$(F;G)(f;g)=G(F(f);F(g))=G(F(f));G(F(g))$	$(F;G)(f;g)=G(F(f);F(g))=G(F(f));G(F(g))$	$(F \circ G)(f \circ g)=G(F(f) \circ F(g))=G(F(f)) \circ G(F(g))$
fong-spivak-seven-sketchesEQ0212	292	■ 1.000	$((F;G);H)(c)=H((F;G)(c))=H(G(F(c)))=(G;H)(F(c))=(F;(G;H))(c)$.	$((F;G);H)(c)=H((F;G)(c))=H(G(F(c)))=(G;H)(F(c))=(F;(G;H))(c)$.	$((F \circ G) \circ H)(c)=H((F \circ G)(c))=H(G(F(c)))=(G \circ H)(F(c))=(F \circ (G \circ H))(c)$.
fong-spivak-seven-sketchesEQ0215	295	■ 1.000	$\left(f_1;f_2\right)^B(g)=g;(f_1;f_2)=(g;f_1);f_2=\left(f_1^B;f_2^B\right)(g)$	$(f_1;f_2)^B(g)=g;(f_1;f_2)=(g;f_1);f_2=\left(f_1^B;f_2^B\right)(g)$	$(f_1 \circ f_2)^B(g)=g \circ (f_1 \circ f_2)=(g \circ f_1) \circ f_2=(f_1^B \circ f_2^B)(g)$
fong-spivak-seven-sketchesEQ0216	296	■ 1.000	$\left(c_1, d_1\right) \rightarrow \left(c_2, d_2\right) \rightarrow \left(c_3, d_3\right) \rightarrow \left(c_4, d_4\right)$.	$(c_1, d_1) \xrightarrow{(f_1, g_1)} (c_2, d_2) \xrightarrow{(f_2, g_2)} (c_3, d_3) \xrightarrow{(f_3, g_3)} (c_4, d_4)$.	$(c_1, d_1) \xrightarrow{(f_1, g_1)} (c_2, d_2) \xrightarrow{(f_2, g_2)} (c_3, d_3) \xrightarrow{(f_3, g_3)} (c_4, d_4)$.
fong-spivak-seven-sketchesEQ0220	298	■ 1.000	$\left(\mathrm{X}^{\mathrm{op}} \times \mathrm{Y}\right)\left((x, y),\left(x', y'\right)\right) \leq \mathcal{V}\left(\Phi(x, y), \Phi\left(x', y'\right)\right)$.	$(\mathcal{X}^{\mathrm{op}} \times y)\left((x, y),\left(x', y'\right)\right) \leq \mathcal{V}\left(\Phi(x, y), \Phi\left(x', y'\right)\right)$.	$(\mathcal{X}^{\mathrm{op}} \times y)\left((x, y),\left(x', y'\right)\right) \leq \mathcal{V}\left(\Phi(x, y), \Phi\left(x', y'\right)\right)$.
fong-spivak-seven-sketchesEQ0231	306	■ 1.000	$\sigma_{m,n}(i):=\begin{cases} i+n & \text{if } i \leq m \\ i-m & \text{if } m+1 \leq i \end{cases}$	$\sigma_{m,n}(i):=\begin{cases} i+n & \text{if } i \leq m \\ i-m & \text{if } m+1 \leq i \end{cases}$	$\sigma_{m,n}(i):=\begin{cases} i+n & \text{if } i \leq m \\ i-m & \text{if } m+1 \leq i \end{cases}$
fong-spivak-seven-sketchesEQ0232	306	■ 1.000	$(f+g)(i):=\begin{cases} f(i) & \text{if } i \leq m \\ g(i-m) & \text{if } m+1 \leq i \end{cases}$	$(f+g)(i):=\begin{cases} f(i) & \text{if } i \leq m \\ g(i-m) & \text{if } m+1 \leq i \end{cases}$	$(f+g)(i):=\begin{cases} f(i) & \text{if } i \leq m \\ g(i-m) & \text{if } m+1 \leq i \end{cases}$
fong-spivak-seven-sketchesEQ0233	308	■ 1.000	$a^{\{0\}}, a^{\{1\}}, a^{\{2\}}, a^{\{3\}}, \ldots, a^{\{2019\}}, \ldots$	$a^0, a^1, a^2, a^3, \ldots, a^{2019}, \ldots$	$a^0, a^1, a^2, a^3, \ldots, a^{2019}, \ldots$
fong-spivak-seven-sketchesEQ0234	308	■ 1.000	$[], \quad [a], \quad [b], \quad [a, a], \quad [a, b], \quad \ldots, \quad [b, a, b, b, a, b, a, a, a, a], \quad \ldots$	$[], \quad [a], \quad [b], \quad [a, a], \quad [a, b], \quad \ldots, \quad [b, a, b, b, a, b, a, a, a, a], \quad \ldots$	$[], \quad [a], \quad [b], \quad [a, a], \quad [a, b], \quad \ldots, \quad [b, a, b, b, a, b, a, a, a, a], \quad \ldots$
fong-spivak-seven-sketchesEQ0235	308	■ 1.000	$G:=\left\{\rho_{m,n}: m \rightarrow n \mid m, n \in \mathbb{N}\right\}, \quad s\left(\rho_{m,n}\right):=m, \quad t\left(\rho_{m,n}\right):=n ;$	$G:=\left\{\rho_{m,n}: m \rightarrow n \mid m, n \in \mathbb{N}\right\}, \quad s\left(\rho_{m,n}\right):=m, \quad t\left(\rho_{m,n}\right):=n ;$	$G:=\left\{\rho_{m,n}: m \rightarrow n \mid m, n \in \mathbb{N}\right\}, \quad s\left(\rho_{m,n}\right):=m, \quad t\left(\rho_{m,n}\right):=n ;$
fong-spivak-seven-sketchesEQ0236	309	■ 1.000	$\left(\begin{array}{llll} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{array}\right)$.	$\left(\begin{array}{llll} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{array}\right)$.	$\left(\begin{array}{llll} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{array}\right)$.

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fong-spivak-seven-sketches EQ0237	309	1.000	$A * B = \left(\begin{array}{ll} 0 & 1 \\ 0 & 0 \end{array} \right) * \left(\begin{array}{ll} 0 & 1 \\ 1 & 0 \end{array} \right) \neq \left(\begin{array}{ll} 0 & 1 \\ 1 & 0 \end{array} \right) * \left(\begin{array}{ll} 0 & 1 \\ 0 & 0 \end{array} \right) = B * A$	$A * B = \left(\begin{array}{ll} 0 & 1 \\ 0 & 0 \end{array} \right) * \left(\begin{array}{ll} 0 & 1 \\ 1 & 0 \end{array} \right) \neq \left(\begin{array}{ll} 0 & 1 \\ 1 & 0 \end{array} \right) * \left(\begin{array}{ll} 0 & 1 \\ 0 & 0 \end{array} \right) = B * A$	$A * B = \left(\begin{array}{ll} 0 & 1 \\ 0 & 0 \end{array} \right) * \left(\begin{array}{ll} 0 & 1 \\ 1 & 0 \end{array} \right) \neq \left(\begin{array}{ll} 0 & 1 \\ 1 & 0 \end{array} \right) * \left(\begin{array}{ll} 0 & 1 \\ 0 & 0 \end{array} \right) = B * A$
fong-spivak-seven-sketches EQ0238	309	1.000	$A+B=\left(\begin{array}{lllllll} 3 & 3 & 1 & 0 & 0 & 0 & 0 \\ 2 & 0 & 4 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 2 & 5 & 6 & 1 \end{array}\right)$	$A+B=\left(\begin{array}{lllllll} 3 & 3 & 1 & 0 & 0 & 0 & 0 \\ 2 & 0 & 4 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 2 & 5 & 6 & 1 \end{array}\right)$	$A+B=\left(\begin{array}{lllllll} 3 & 3 & 1 & 0 & 0 & 0 & 0 \\ 2 & 0 & 4 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 2 & 5 & 6 & 1 \end{array}\right)$
fong-spivak-seven-sketches EQ0239	310	1.000	$\left(\begin{array}{lll} 1 & 1 & 1 \end{array}\right)$	$\left(\begin{array}{lll} 1 & 1 & 1 \end{array}\right)$	$\left(\begin{array}{lll} 1 & 1 & 1 \end{array}\right)$
fong-spivak-seven-sketches EQ0240	310	1.000	$\left(\begin{array}{lll} 1 & 1 & 1 \end{array}\right)$	$\left(\begin{array}{lll} 1 & 1 & 1 \end{array}\right)$	$\left(\begin{array}{lll} 1 & 1 & 1 \end{array}\right)$
fong-spivak-seven-sketches EQ0241	310	1.000	$g_1:=c_n+\cdots+c_n:m\rightarrow(m\times n),$	$g_1:=c_n+\cdots+c_n:m\rightarrow(m\times n),$	$g_1:=c_n+\cdots+c_n:m\rightarrow(m\times n),$
fong-spivak-seven-sketches EQ0243	310	1.000	$\begin{aligned} g_2:=& s_{M(1,1)}+\cdots+s_{M(1,n)} \\ & +s_{M(2,1)}+\cdots+s_{M(2,n)} \\ & +\cdots \\ & +s_{M(m,1)}+\cdots+s_{M(m,n)}:(m\times n)\rightarrow(m\times n), \end{aligned}$	$g_2:=s_{M(1,1)}+\cdots+s_{M(1,n)}+s_{M(2,1)}+\cdots+s_{M(2,n)}+\cdots+s_{M(m,1)}+\cdots+s_{M(m,n)}:(m\times n)\rightarrow(m\times n),$	$g_2:=s_{M(1,1)}+\cdots+s_{M(1,n)}+s_{M(2,1)}+\cdots+s_{M(2,n)}+\cdots+s_{M(m,1)}+\cdots+s_{M(m,n)}:(m\times n)\rightarrow(m\times n),$
fong-spivak-seven-sketches EQ0244	310	1.000	$g_4:=a_m+\cdots+a_m:(m\times n)\rightarrow n,$	$g_4:=a_m+\cdots+a_m:(m\times n)\rightarrow n,$	$g_4:=a_m+\cdots+a_m:(m\times n)\rightarrow n,$
fong-spivak-seven-sketches EQ0246	312	1.000	$\mu:=\left(\begin{array}{ll} 1 & 0 \\ 0 & 1 \\ 1 & 0 \\ 0 & 1 \end{array}\right)$	$\mu:=\left(\begin{array}{ll} 1 & 0 \\ 0 & 1 \\ 1 & 0 \\ 0 & 1 \end{array}\right)$	$\mu:=\left(\begin{array}{ll} 1 & 0 \\ 0 & 1 \\ 1 & 0 \\ 0 & 1 \end{array}\right)$
fong-spivak-seven-sketches EQ0247	313	1.000	$U(\mu)\left(\left(a_1,\ldots,a_m\right),\left(b_1,\ldots,b_m\right)\right):=\left(a_1+b_1,\ldots,a_m+b_m\right).$	$U(\mu)\left(\left(a_1,\ldots,a_m\right),\left(b_1,\ldots,b_m\right)\right):=\left(a_1+b_1,\ldots,a_m+b_m\right).$	$U(\mu)\left(\left(a_1,\ldots,a_m\right),\left(b_1,\ldots,b_m\right)\right):=\left(a_1+b_1,\ldots,a_m+b_m\right).$
fong-spivak-seven-sketches EQ0248	313	1.000	$B+C:=\left\{\left(w,y,x,z\right)\in R^{m+p}\times R^{n+q}\mid\left(w,x\right)\in B\text{ and }\left(y,z\right)\in C\right\}.$	$B+C:=\left\{\left(w,y,x,z\right)\in R^{m+p}\times R^{n+q}\mid\left(w,x\right)\in B\text{ and }\left(y,z\right)\in C\right\}.$	$B+C:=\left\{\left(w,y,x,z\right)\in R^{m+p}\times R^{n+q}\mid\left(w,x\right)\in B\text{ and }\left(y,z\right)\in C\right\}.$
fong-spivak-seven-sketches EQ0249	313	1.000	$\begin{aligned} \mathrm{B}(g)&=\left\{\left(x,z\right)\in R^m\times R^n\mid S(g)(x)=z\right\} \\ \mathrm{B}(h^{\mathrm{op}})&=\left\{\left(z,y\right)\in R^n\times R^\ell\mid z=S(h)(y)\right\} \end{aligned}$	$\begin{aligned} \mathrm{B}(g)&=\left\{\left(x,z\right)\in R^m\times R^n\mid S(g)(x)=z\right\} \\ \mathrm{B}(h^{\mathrm{op}})&=\left\{\left(z,y\right)\in R^n\times R^\ell\mid z=S(h)(y)\right\} \end{aligned}$	$\begin{aligned} \mathrm{B}(g)&=\left\{\left(x,z\right)\in R^m\times R^n\mid S(g)(x)=z\right\} \\ \mathrm{B}(h^{\mathrm{op}})&=\left\{\left(z,y\right)\in R^n\times R^\ell\mid z=S(h)(y)\right\} \end{aligned}$
fong-spivak-seven-sketches EQ0254	313	1.000	$\mathrm{B}\left(\left(g^{\mathrm{op}}\right);h\right)=\left\{\left(S(g)(x),S(h)(x)\right)\mid x\in R^m\right\}.$	$\mathrm{B}\left(\left(g^{\mathrm{op}}\right);h\right)=\left\{\left(S(g)(x),S(h)(x)\right)\mid x\in R^m\right\}.$	$\mathrm{B}\left(\left(g^{\mathrm{op}}\right);h\right)=\left\{\left(S(g)(x),S(h)(x)\right)\mid x\in R^m\right\}.$
fong-spivak-seven-sketches EQ0255	315	1.000	$f(m)=f(\underbrace{1+1+\cdots+1}_m)=\underbrace{f(1)+f(1)+\cdots+f(1)}_m=\underbrace{1_R+1_R+\cdots+1_R}_m.$	$f(m)=f(\underbrace{1+1+\cdots+1}_m)=\underbrace{f(1)+f(1)+\cdots+f(1)}_m=\underbrace{1_R+1_R+\cdots+1_R}_m.$	$f(m)=f(\underbrace{1+1+\cdots+1}_m)=\underbrace{f(1)+f(1)+\cdots+f(1)}_m=\underbrace{1_R+1_R+\cdots+1_R}_m.$

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fong-spivak-seven-sketches EQ0261 (A.3)	327	■ 1.000	$\begin{array}{cc c c} P & Q & P \wedge Q & P = (P \wedge Q) \\ \hline \text{true} & \text{true} & \text{true} & \text{true} \\ \text{true} & \text{false} & \text{false} & \text{false} \\ \text{false} & \text{true} & \text{false} & \text{true} \\ \text{false} & \text{false} & \text{false} & \text{true} \end{array}$	$U = \bigcup_{i \in I} \{x \in \mathbb{R} \mid a_i < x < b_i\}$	$U = \bigcup_{j \in J} \{x \in \mathbb{R} \mid a_j < x < b_j\} = \bigcup_{j \in J} (o_{[a_j, b_j]} \cap \mathbb{R}) = \left(\bigcup_{j \in J} o_{[a_j, b_j]} \right) \cap \mathbb{R}$
fong-spivak-seven-sketches EQ0263	333	■ 1.000	$U = \bigcup_{i \in I} \{x \in \mathbb{R} \mid a_i < x < b_i\}$	$U = \bigcup_{j \in J} \{x \in \mathbb{R} \mid a_j < x < b_j\} = \bigcup_{j \in J} (o_{[a_j, b_j]} \cap \mathbb{R}) = \left(\bigcup_{j \in J} o_{[a_j, b_j]} \right) \cap \mathbb{R}$	$U = \bigcup_{j \in J} \{x \in \mathbb{R} \mid a_j < x < b_j\} = \bigcup_{j \in J} (o_{[a_j, b_j]} \cap \mathbb{R}) = \left(\bigcup_{j \in J} o_{[a_j, b_j]} \right) \cap \mathbb{R}$
fong-spivak-seven-sketches EQ0264	333	■ 1.000	$U = \bigcup_{j \in J} \{x \in \mathbb{R} \mid a_j < x < b_j\} = \bigcup_{j \in J} (o_{[a_j, b_j]} \cap \mathbb{R}) = \left(\bigcup_{j \in J} o_{[a_j, b_j]} \right) \cap \mathbb{R}$	$U = \bigcup_{j \in J} \{x \in \mathbb{R} \mid a_j < x < b_j\} = \bigcup_{j \in J} (o_{[a_j, b_j]} \cap \mathbb{R}) = \left(\bigcup_{j \in J} o_{[a_j, b_j]} \right) \cap \mathbb{R}$	$U = \bigcup_{j \in J} \{x \in \mathbb{R} \mid a_j < x < b_j\} = \bigcup_{j \in J} (o_{[a_j, b_j]} \cap \mathbb{R}) = \left(\bigcup_{j \in J} o_{[a_j, b_j]} \right) \cap \mathbb{R}$